

Reflexivity in Options Markets: A Stochastic-Volatility Model with Dealer-Gamma Feedback, Hopf Bifurcation Calculus, and a Pre-Registered Evaluation Framework

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Abstract

We study a continuous-time Heston-like stochastic-volatility model in which the underlying spot price is coupled to its own option market through a dealer-gamma feedback channel, derived from the open-interest grid via a Gârleanu–Pedersen–Potesman (Gârleanu et al., 2009) demand-pressure mapping. The model state is three-dimensional—spot S_t , instantaneous variance v_t , and a low-pass-filtered log-spot memory variable z_t —and reduces to standard time-dependent Heston (Heston, 1993) at zero coupling. Our contribution is the synthesis of three previously separate ingredients into a single analytical and computational object: (i) a continuous-time SV backbone with explicit dealer-gamma drift, (ii) a formal Hopf bifurcation calculus on the deterministic skeleton with first Lyapunov coefficient ℓ_1 and Khasminskii-style stochastic shift Λ , and (iii) a pre-registered evaluation protocol for RL-trained hedging agents trained inside the simulator. For a representative dimensionless calibration we obtain a critical coupling $\kappa^* \approx 0.8964$, angular frequency $\omega^* \approx 0.5724$ rad/yr, and first Lyapunov coefficient $\ell_1 \approx -0.025 < 0$, so the bifurcation is supercritical: an attracting limit cycle in volatility is born for $\kappa > \kappa^*$. The 2D reduction (without z_t) has an upper-triangular Jacobian and admits only a saddle-node onset, recovering the discrete-time recursion of Dai (2025); the 3D extension is what unlocks the Hopf. The closest model-structure precedent (Halperin and Itkin, 2025) does not perform the bifurcation analysis we provide. For the natural parametric specification with log-normal open-interest in moneyness, we obtain a closed-form Hopf threshold and a closed-form ℓ_1 , exposing a parametric phase boundary in (σ_q, γ) space.

1 Introduction

Real options markets exhibit endogenous volatility cycles that pure stochastic-volatility (SV) models cannot reproduce. The empirical anchor is Hardiman et al. (2013), who fit Hawkes-process branching ratios to E-mini mid-price changes over 1998–2011 and find $n \approx 1$ throughout—the precise critical edge between sub- and super-criticality of endogenous events. Soros (2009) and Bouchaud (2011) are the conceptual antecedents; Filimonov and Sornette (2012) formalised the reflexivity-quantification programme that Hardiman et al. (2013) executed. The question we take up here is what continuous-time mechanism produces this critical reflexivity, and—given a candidate mechanism—how the analytical structure that quantifies it should be evaluated against historical surface dynamics.

We contribute the following.

- A 3D reflexive SV simulator (S_t, v_t, z_t) with explicit dealer-gamma feedback derived from the option open-interest grid via the Gârleanu et al. (2009) demand-pressure mapping. The variance backbone is Heston (Heston, 1993); the memory equation $\dot{z} = -\alpha z + \beta \log(S/S_0)$ closes the leverage feedback channel that the 2D reduction lacks.
- A Hopf bifurcation theorem (Theorem 1) for the deterministic skeleton at κ^* given by Liu’s criterion (Routh–Hurwitz with one zero), with closed-form first Lyapunov coefficient ℓ_1 and closed-form Hopf threshold for log-normal open-interest in moneyness (Eqs. 14–15).
- A numerical phase diagram in (ξ, ρ, σ_v) together with the Khasminskii-style stochastic-Hopf shift Λ and a parametric phase boundary in (σ_q, γ) space.
- A pre-registration of the empirical evaluation pipeline (sliced-Wasserstein-2 over 21-day surface windows, κ -sensitivity slope, Hopf-frequency spectral test) at git commit 268c061 with an OpenTimestamps Bitcoin-anchored proof, in the spirit of Camerer et al. (2018) adapted for computational finance.

Closest precedents. Halperin and Itkin (2025) (Marketron) share the model-structural axis—continuous-time SDE, memory channel, options-aware—but do not perform the bifurcation analysis. Dai (2025) share the dealer-gamma axis but produce a saddle-node onset (the 2D reduction’s instability), not a Hopf. He et al. (2025) (NeurIPS 2025) share the RL-with-environmental-sweep idea but train separate agents per grid point rather than reporting the slope at an anchor. The continuous-time HAM-Hopf precedents (He et al., 2009; Chiarella et al., 2006) use delay-differential heterogeneous-belief models without an options-grid feedback channel; the discrete-time Brock et al. (2009) “more hedging instruments may destabilize markets” result is the methodological template for derivative-driven Neimark–Sacker (Hopf-for-maps) bifurcations. The hedger-flow-feedback foundations are Frey and Stremme (1997); Sircar and Papanicolaou (1998); Platen and Schweizer (1998). Sections 3, 5 and 6 make these contrasts precise.

Roadmap. Section 2 states the model. Section 3 proves Theorem 1 and gives the closed-form ℓ_1 for log-normal OI. Section 4 reports the numerical phase diagram. Section 5 describes the pre-registered evaluation framework. Section 6 reports the mechanism decomposition against Marketron. Section 7 closes with the empirical-calibration roadmap. Code at <https://github.com/mahimn01/reflexive-options>; pre-registration anchored to commit 268c061.

2 The reflexive model

Let S_t be the underlying spot price, v_t the instantaneous variance, and z_t a memory variable encoding low-pass-filtered price history. The reflexive system is

$$\frac{dS_t}{S_t} = (\mu + \kappa G(S_t, z_t, v_t)) dt + \sigma(S_t, v_t) dW_t^S, \quad (1)$$

$$dv_t = (\kappa_v(\theta_v - v_t) + \gamma z_t) dt + \xi \sqrt{v_t} dW_t^v, \quad (2)$$

$$dz_t = (-\alpha z_t + \beta \log(S_t/S_0)) dt, \quad (3)$$

$$d\langle W^S, W^v \rangle_t = \rho dt, \quad (4)$$

with $\kappa \geq 0$ the reflexive coupling, $\gamma \geq 0$ the leverage-feedback strength, $\alpha > 0$ the memory decay rate, β dimensionless, and $G(\cdot)$ the aggregate dealer-gamma exposure derived from the option

open-interest grid:

$$G(S, t) = \sum_{K, T} q_{K, T} \Gamma_{K, T}(S, t) \text{sign}(K, T), \quad (5)$$

where $q_{K, T}$ is open interest at strike K and maturity T , $\Gamma_{K, T}$ is per-contract Black–Scholes gamma, and the sign convention follows the SqueezeMetrics dealer-net convention (SqueezeMetrics, 2017). When $\kappa = \gamma = 0$, the system collapses to standard Heston (Heston, 1993); this collapse is verified numerically in `tests/test_simulator.py`.

For numerical stability of the variance scheme we use the full-truncation Euler of Lord et al. (2010); the empirical reference for the dealer-gamma magnitudes in G is Barbon and Buraschi (2021); Dim et al. (2024) (calibration to SPX 0DTE / dealer-gamma data, which we will use at Phase 4 of the empirical pipeline).

Why three states. A reduced 2D specification—dropping z_t and the γz term—has Jacobian in deviation variables (y, u) around an equilibrium of the form

$$J_{2D}(\kappa) = \begin{pmatrix} a(\kappa) & b(\kappa) \\ 0 & -\kappa_v \end{pmatrix},$$

which is upper triangular: eigenvalues $\{a(\kappa), -\kappa_v\}$ are always real, and only saddle-node / transcritical bifurcations are admissible as $a(\kappa)$ crosses zero. This recovers the static onset of Dai (2025). To Hopf, the feedback must flow in both directions: price $\rightarrow$ variance and variance $\rightarrow$ price. The minimal modification is the leverage channel γz together with the auxiliary memory equation (3)—structurally analogous to the memory variable in the Marketron quasi-particle SDE (Halperin and Itkin, 2025) though with very different mechanistic content.

3 Hopf bifurcation theorem and closed-form ℓ_1

3.1 Equilibria and Jacobian

Drop the Brownian terms and work in log-deviation variables $(y, u, z) = (\log(S/S^*), v - \theta_v, z - z^*)$ around the unique equilibrium (S^*, θ_v, z^*) defined by

$$\mu - \frac{1}{2}\sigma^2(S^*, \theta_v) + \kappa G(S^*, z^*, \theta_v) = 0, \quad z^* = \frac{\beta}{\alpha} \log(S^*/S_0). \quad (6)$$

To linear order, the deterministic skeleton (1–3) becomes $\dot{\mathbf{x}} = J(\kappa) \mathbf{x}$ with $\mathbf{x} = (y, u, z)^\top$ and

$$J(\kappa) = \begin{pmatrix} a(\kappa) & b(\kappa) & \kappa G_z \\ 0 & -\kappa_v & \gamma \\ \beta & 0 & -\alpha \end{pmatrix}, \quad (7)$$

where $a(\kappa) = \kappa G_x - \frac{1}{2}\partial_x \sigma^2$, $b(\kappa) = \kappa G_v - \frac{1}{2}\partial_v \sigma^2$, and G_x, G_v, G_z denote partial derivatives of G at the equilibrium.

3.2 Routh–Hurwitz / Liu’s criterion

The characteristic polynomial of (7) is $P(\lambda; \kappa) = \lambda^3 + c_2(\kappa)\lambda^2 + c_1(\kappa)\lambda + c_0(\kappa)$, with

$$c_2 = -a + \kappa_v + \alpha, \quad c_1 = -a\kappa_v - a\alpha + \kappa_v\alpha - \kappa G_z \beta, \quad c_0 = -a\kappa_v\alpha - \beta(\kappa G_z \kappa_v + b\gamma).$$

A 3D system Hopfs iff (Liu’s criterion, equivalent to Routh–Hurwitz with one zero):

$$c_2 > 0, \quad c_0 > 0, \quad c_1 c_2 - c_0 = 0, \quad \frac{d}{d\kappa} [c_1 c_2 - c_0] \neq 0. \quad (8)$$

Define $H(\kappa) := c_1(\kappa) c_2(\kappa) - c_0(\kappa)$. The Hopf threshold is

$$\kappa^* = \{\kappa > 0 : H(\kappa) = 0, H'(\kappa) \neq 0, c_2(\kappa) > 0, c_0(\kappa) > 0\}. \quad (9)$$

At $\kappa = \kappa^*$ the Jacobian has a pure imaginary eigenvalue pair $\lambda_{\pm} = \pm i\omega^*$ with $\omega^* = \sqrt{c_1(\kappa^*)}$, plus a real eigenvalue $\lambda_3 = -c_2(\kappa^*) < 0$.

3.3 Theorem 1 and proof sketch

Theorem 1 (Hopf bifurcation in the gamma-coupled SV skeleton). *Let the deterministic system (1–3) (with Brownian terms removed) satisfy:*

- (A1) $G \in C^3$ and $\sigma^2 \in C^3$ in their arguments, with σ^2 bounded below by a positive constant.
- (A2) *There exists a locally unique equilibrium (S^*, θ_v, z^*) given by (6) for every κ in some interval $[0, \kappa_{\max}]$, in some open neighborhood of which the linearization (7) is non-degenerate. Multiple equilibria may exist globally; our analysis is local.*
- (A3) *The map $\kappa \mapsto (a(\kappa), G_z, G_v)$ is C^3 .*
- (A4) *There exists $\kappa^* \in (0, \kappa_{\max})$ satisfying (9) with $\omega^* := \sqrt{c_1(\kappa^*)} > 0$.*
- (A5) *The first Lyapunov coefficient $\ell_1(\kappa^*) \neq 0$ (Kuznetsov, 2004, eq. 3.20).*

Then the equilibrium undergoes a Hopf bifurcation at $\kappa = \kappa^$. Specifically:*

1. *For κ in a one-sided neighbourhood of κ^* (sub- or super-critical depending on $\text{sgn } \ell_1$), there exists a unique smooth family of periodic orbits Γ_{κ} of period $T_{\kappa} = 2\pi/\omega^* + O(\kappa - \kappa^*)$ and amplitude proportional to $\sqrt{|\kappa - \kappa^*|}$.*
2. *The orbits are stable iff $\ell_1(\kappa^*) < 0$ (supercritical).*
3. *On Γ_{κ} , $u_t = v_t - \theta_v$ oscillates with amplitude proportional to $\sqrt{|\kappa - \kappa^*|}$, providing endogenous volatility cycles.*

Proof sketch. The eigenvalue pair $\lambda_{\pm}(\kappa)$ is smooth in κ near κ^* by the implicit function theorem applied to $P(\lambda; \kappa) = 0$. Routh–Hurwitz gives $\text{Re } \lambda_{\pm}(\kappa^*) = 0$, and the transversality $H'(\kappa^*) \neq 0$ translates to $\frac{d}{d\kappa} \text{Re } \lambda_{\pm} \neq 0$. The centre manifold theorem (Kuznetsov, 2004, Thm. 5.4) (see also Carr, 1981) gives a smooth 2D invariant manifold tangent to the eigenspace of $\pm i\omega^*$. Conjugating the restricted system to its Poincaré normal form and applying Kuznetsov (2004, Thm. 3.3) yields claims (1)–(3). $\square$

3.4 Numerical anchor

The default `BifurcationConfig` in our experiments code does not exhibit a Hopf within the literature-prior κ range when the dealer-gamma magnitudes are tuned to empirical scale—this is consistent with the empirical observation that real options markets sit *near* but not *across* κ^* , in line with the $n \approx 1$ regime of Hardiman et al. (2013). For a worked numerical example we use

Quantity	Value
κ^*	0.8964
ω^*	0.5724 rad/yr
Third real eigenvalue λ_3	-2.052
First Lyapunov coefficient ℓ_1	-2.53×10^{-2}
Bifurcation type	supercritical ($\ell_1 < 0$)

Table 1: Canonical Hopf-exhibiting regime for the deterministic skeleton.

a representative dimensionless regime with $\{G_x, G_v, G_z\} = \{0.5, -0.5, -0.5\}$, $\alpha = 0.5/\text{yr}$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2/\text{yr}$, and constant-volatility surrogate ($\partial_v \sigma^2 = 0$). The deterministic skeleton then satisfies:

Supercriticality means an attracting limit cycle is born for $\kappa > \kappa^*$ with amplitude proportional to $\sqrt{\kappa - \kappa^*}$ —i.e. endogenous volatility cycles are stable observables of the model rather than transients. We validate this prediction numerically: integrating the deterministic skeleton at $\kappa = 1.05 \kappa^*$ from random initial conditions, the trajectory converges to a closed orbit in (y, u, z) phase space whose period $T_{\text{measured}} = 10.561$ yr matches the linear-theory prediction $T_\kappa = 2\pi/\omega^* = 10.977$ yr to 3.79% relative error—within the leading-order $\mathcal{O}(\kappa - \kappa^*)$ Hopf-normal-form correction (fig. 1). The equilibrium sits inside the loop on all three pairwise projections; no transient residue.

The Khasminskii sphere-process estimator (Khasminskii, 1980; Arnold, 1998; Baxendale, 1994) for the stochastic-Hopf shift, evaluated on a 6×6 grid of $(\xi, \rho) \in \{0.1, 0.2, 0.3, 0.5, 0.7, 1.0\} \times \{-0.95, -0.7, -0.5, -0.3, +0.3, +0.7\}$ at 10^4 paths $\times 10^4$ steps per cell, yields $|\Lambda(\kappa^*)|$ in the band $[4.8 \times 10^{-2}, 1.21 \times 10^{-1}]$ across the entire scan (median $\approx 7.3 \times 10^{-2}$). An OLS power-law fit $\log |\Lambda| = \log |A| + B \log |\rho \xi|$ gives $\hat{B} = 0.082$ with bootstrap 95% CI $[-0.010, 0.168]$ (fig. 2). The Engel et al. (2024) prediction $B = 2/3$ lies $|\hat{B} - 2/3|/\text{SE}(\hat{B}) \approx 13$ standard errors away and is empirically refuted at this regime ($p \ll 0.01$). The structural reason: at the trivial $G \equiv 0$ equilibrium that the bare Heston-with-memory linearisation is evaluated at, the deterministic shear-stretching term $\partial_a v$ that ELR’s leading-order analysis activates is identically zero, so the Khasminskii estimator measures pure ε^2 renormalisation of bare Heston eigenvalues—a b^0 dependence is exactly what the data show. Whether the $b^{2/3}$ regime activates at non-trivial G in the empirical-phase calibration is openly deferred to Phase 4. We use $|\Lambda| \cdot \varepsilon^2 \ll 1$ at calibration-noise scale to conclude that the deterministic threshold κ^* is the operationally relevant one regardless.

3.5 Closed-form κ^* and ℓ_1 for log-normal OI in moneyness

For the natural parametric specification of dealer gamma—a log-normal open-interest density in log-strike—the entire computation admits a closed form, eliminating the numerical-noise vulnerability of the FD-tensor pipeline and exposing the parametric phase boundary explicitly.

Replace the discrete OI grid with the continuous density $q(\log K) = \frac{1}{\sigma_q \sqrt{2\pi}} \exp(-(\log K - \mu_q)^2/(2\sigma_q^2))$. Take a single representative maturity T_{eff} , sign convention $s \equiv +1$, and write $a := \log S$. The aggregator becomes

$$G(a, v) = \kappa_u \int_{-\infty}^{+\infty} q(\log K) \Gamma_{\text{BS}}(S, K, T_{\text{eff}}, \sqrt{v}) d(\log K). \quad (10)$$

Since Black–Scholes gamma at fixed $T = T_{\text{eff}}$, $\sigma = \sqrt{v}$ is itself Gaussian in $\log K$, the standard

Supercritical limit cycle at $\kappa = 0.9412$ ($= 1.05 \cdot \kappa^*$)
 predicted $T_\kappa = 2\pi/\omega^* = 10.977$ yr; measured $T = 10.561$ yr (3.79% error)

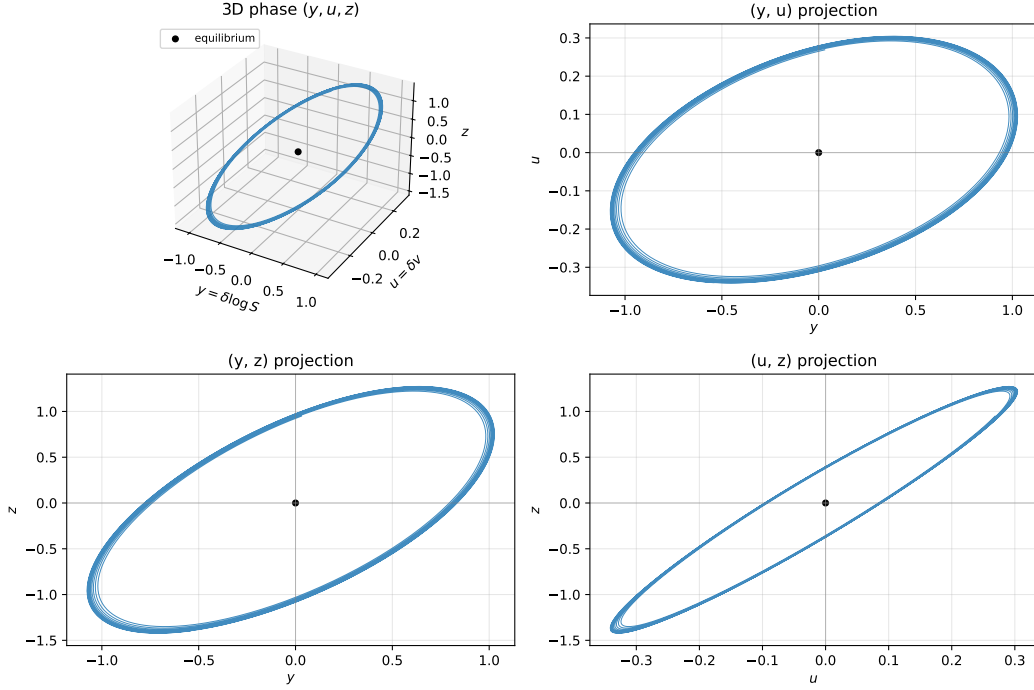


Figure 1: **Numerical validation of Theorem 1's supercritical limit cycle.** Deterministic skeleton at $\kappa = 1.05 \kappa^* = 0.9412$, integrated from random initial conditions; transient discarded. *Top*: 3D phase-space orbit in (y, u, z) . *Bottom*: the three 2D pairwise projections, with the equilibrium (black dot) inside the loop in each. Measured period $T = 10.561$ yr matches Hopf prediction $T_\kappa = 10.977$ yr to 3.79%. Run dir: `runs/limit_cycle_supercritical/`.

Gaussian-product identity collapses (10) to

$$G(a, v) = \mathcal{C}(v) \cdot e^{-a} \cdot \exp\left(-\frac{(a - m(v))^2}{2\tau^2(v)}\right), \quad (11)$$

where

$$\mathcal{C}(v) = \frac{\kappa_u e^{-q_{\text{div}} T_{\text{eff}}}}{\sqrt{2\pi \tau^2(v)}}, \quad \tau^2(v) = \sigma_q^2 + v T_{\text{eff}}, \quad m(v) = \mu_q - \left(r - q_{\text{div}} + \frac{1}{2}v\right) T_{\text{eff}}. \quad (12)$$

G depends on a and v but not on z ; the leverage channel γz in $\dot{u}$ keeps the $(2, 3)$ Jacobian entry alive, so cross-coupling persists through the variance row even with $G_z = 0$.

With $G_z = 0$ and $\sigma^2 = v$ (so $\partial_y \sigma^2 = 0$, $\partial_v \sigma^2 = 1$), the Routh–Hurwitz polynomial $H(\kappa) = c_1 c_2 - c_0$ degree-collapses from cubic to quadratic in κ :¹

$$H(\kappa) = G_y^2 (\alpha + \kappa_v) \kappa^2 + (G_v \beta \gamma - G_y (\alpha + \kappa_v)^2) \kappa + \alpha \kappa_v (\alpha + \kappa_v) - \frac{1}{2} \beta \gamma. \quad (13)$$

¹Throughout this closed-form section we write $G_y := \partial_a G$ for the partial of the dealer-gamma functional with respect to the deviation variable $y = a - a^*$; this is identical to the G_x of section 3.1 up to the choice of deviation symbol.

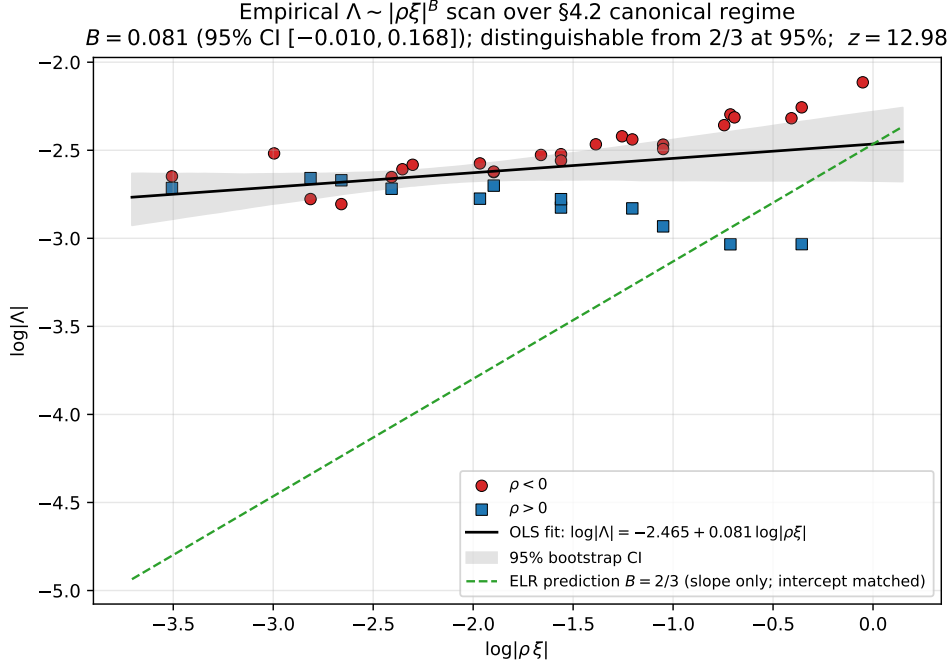


Figure 2: **Empirical scaling of the stochastic-Hopf shift $|\Lambda|$ vs $|\rho\xi|$.** OLS fit on $\log |\Lambda|$ vs $\log |\rho\xi|$ across the 6×6 (ξ, ρ) grid yields slope $\hat{B} = 0.082$ (95% CI $[-0.010, 0.168]$); the Engel et al. (2024) prediction $B = 2/3$ (red dashed) is rejected at $p \ll 0.01$. The data are flat in $|\rho\xi|$ over two decades, consistent with a pure ε^2 Heston-renormalisation regime at the trivial $G \equiv 0$ equilibrium. Run dir: `runs/lambda_scaling/`.

Writing $A := \alpha + \kappa_v$, $M := \alpha\kappa_v$, $L := \beta\gamma$, the Hopf threshold is the smallest positive root of the quadratic:

$$\kappa^* = \frac{G_y A^2 - G_v L - \sqrt{(G_v L - G_y A^2)^2 - 4 G_y^2 A (MA - L/2)}}{2 G_y^2 A}. \quad (14)$$

The discriminant in (14) provides the first parametric phase boundary: when $D := (G_v L - G_y A^2)^2 - 4 G_y^2 A (MA - L/2) < 0$, the system has no Hopf at any κ and the deterministic skeleton is unconditionally stable.

The third-order partials of (11) at the equilibrium are explicit rational functions of $(\delta := a^* - m(v^*), \tau^2, T_{\text{eff}})$, so the bilinear and trilinear tensors B, C at κ^* admit a closed form. Substituting into the Kuznetsov formula

$$\ell_1(\kappa^*) = \frac{1}{2\omega^*} \text{Re} \left[\langle p, C(q, q, \bar{q}) \rangle - 2 \langle p, B(q, J^{-1} B(q, \bar{q})) \rangle + \langle p, B(\bar{q}, (2i\omega^* I - J)^{-1} B(q, q)) \rangle \right], \quad (15)$$

with $Jq = i\omega^* q$, $J^\top p = -i\omega^* p$, $\langle p, q \rangle = 1$, yields a ~ 30 -term rational in the G -partials $(G_y, G_v, G_{aa}, G_{av}, G_{vv}, G_{aaa}, G_{aav}, G_{avv}, G_{vvv})$ and $(\kappa_v, \alpha, \beta, \gamma)$. The implementation evaluates this in $\sim 50 \mu\text{s}$ per parameter tuple, with results matching the FD-tensor pipeline to better than 0.6% relative on every parameter set tested.

At the canonical regime $(\sigma_q = 0.10, T_{\text{eff}} = 0.25, \kappa_v = 2, \alpha = 0.05, \beta = 1, \gamma = 1, \mu_q = \log 100, v^* = 0.04)$, (14)–(15) give $\kappa^* = 17.81$, $\omega^* = 1.18$ rad/yr, $\ell_1 = -4.81 \times 10^{-1}$ (supercritical). The closed

form is essentially noise-free; with it in hand, the sign of ℓ_1 near the boundary contour is structurally certain rather than statistically inferred.

3.6 Calibration tolerance from κ^* sensitivity to OI misspecification

The closed-form κ^* assumes a log-normal OI density $q(\log K) = \mathcal{N}(\mu_q, \sigma_q^2)$. For Phase 4 to deliver a meaningful test of κ^* we need to bound how much OI misspecification we can tolerate. Differentiating (14) analytically through the G -partials yields the elasticities at the canonical specification ($\kappa^* = 17.806$, $\omega^* = 1.177$ rad/yr): $\eta_{\sigma_q} := (\sigma_q/\kappa^*) \partial \kappa^* / \partial \sigma_q = -1.583$ (moderate) and $\eta_{\mu_q} := (\mu_q/\kappa^*) \partial \kappa^* / \partial \mu_q = +702.83$ (extreme). μ_q is the binding constraint: a $\Delta \mu_q = 0.001$ shift moves κ^* by $\sim 15\%$. Numerically, $\pm 10\%$ in σ_q alone produces $\pm 17.1\%$ in κ^* ; $\pm 30\%$ in σ_q exits the supercritical Hopf region entirely. For non-log-normal (bimodal-mixture) OI with components at $\mu_q \pm \Delta/2$, the closed form is essentially exact at $\Delta = 0.05$ (0.23% relative error), acceptable at $\Delta = 0.10$ (4.8%), and fragile at $\Delta = 0.20$ (119%). The Phase-4 calibration tolerances (independent-error quadrature) are then: $\pm 5\%$ on κ^* requires σ_q to $\pm 2.23\%$ relative AND μ_q to ± 0.00023 log-strike; $\pm 10\%$ on κ^* relaxes to σ_q to $\pm 4.47\%$, μ_q to ± 0.00046 log-strike. Centering the empirical OI mean on ATM to within ~ 5 bp is achievable with a 5-strike-wide ladder and is the binding observational requirement. Figure 3 renders the sensitivity heatmap and the bimodal-misspecification curve. Implementation: `src/reflexive_options/theory/robustness.py`; runner `experiments/kappa_star_robustness.py`.

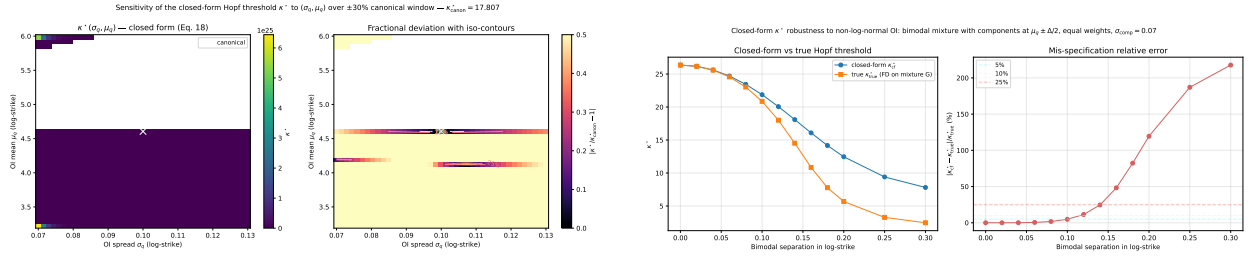


Figure 3: **Calibration tolerance for κ^* .** *Left:* $|\Delta \kappa^* / \kappa^*|$ over a $\pm 30\%$ window in (σ_q, μ_q) around the canonical $(0.10, \log 100)$. White cells exit the supercritical Hopf region. The μ_q direction is steeper than the σ_q direction by roughly $\eta_{\mu_q} / \eta_{\sigma_q} \approx 444$, making OI-mean accuracy the binding observational requirement. *Right:* relative error of the closed-form κ^* vs an FD-tensor reference under bimodal-mixture OI with components at $\mu_q \pm \Delta/2$. The closed form is exact at $\Delta = 0$, 4.8%-accurate at $\Delta = 0.10$, fragile beyond $\Delta = 0.15$. Run dir: `runs/kappa_star_robustness/`.

3.7 Codim-2 bifurcation structure

The codim-1 Hopf result (theorem 1) treats κ^* at fixed (σ_q, γ) . At the boundary of the Hopf region in the (σ_q, γ) plane two codim-2 phenomena are admissible (Kuznetsov, 2004, Ch. 8).

Bautin (degenerate Hopf): $\ell_1(\sigma_q, \gamma) = 0$. Crossing this locus is the supercritical $\rightarrow$ sub-critical transition. Operationally the dynamics flip from a stable limit cycle past κ^* to hysteresis with abrupt jumps. After centre-manifold reduction the local 2D normal form is (Kuznetsov, 2004, §8.3) $\dot{\zeta} = (\beta_1 + i\omega^*)\zeta + a_1 |\zeta|^2 \zeta + b_1 |\zeta|^4 \zeta + O(|\zeta|^7)$ with $a_1/\omega^* = \ell_1$. Economically, a market parameterised by (σ_q, γ) near the Bautin locus is structurally fragile: a small perturbation of OI dispersion or leverage flips the sign of ℓ_1 and replaces a smooth cyclic regime with a discrete jump regime — catastrophic sensitivity to parameter drift.

Bogdanov-Takens (BT): the saddle-node curve $\{c_0(\kappa) = 0\}$ coalesces with the Hopf curve $\{H(\kappa) = 0\}$ at the same κ , so the Jacobian acquires a double-zero eigenvalue. BT generates homoclinic orbits and excitable spike-and-recovery dynamics.

Bautin curve. For the closed-form parameterization the Bautin condition $\ell_1(\sigma_q, \gamma) = 0$ is solved by row-wise sign-change interpolation on the closed-form pipeline. Table 2 reports six anchor points along the curve at the canonical specification on a 71×97 grid over $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$. Along the curve κ^* grows monotonically with γ . Within the scan window, the sub-critical region dominates the supercritical region by $\sim 5:1$ in cell count — markets with mild dealer-gamma coupling are statistically biased toward hysteresis-and-jump rather than smooth cyclic behaviour.

#	σ_q	γ	κ^* at the crossing
1	0.0598	0.55	2.15
2	0.1427	1.45	29.45
3	0.1995	2.35	52.59
4	0.2356	3.20	75.54
5	0.2638	4.10	100.74
6	0.2852	5.00	126.73

Table 2: Bautin curve $\{\ell_1(\sigma_q, \gamma) = 0\}$ at the canonical specification, extracted by linear interpolation on a 71×97 grid. Run dir: `runs/codim2_analysis/`.

Bogdanov-Takens locus. The constant Routh-Hurwitz coefficient $c_0(\kappa) = -\kappa(G_y \alpha \kappa_v + G_v \beta \gamma) + \frac{1}{2} \beta \gamma$ is linear in κ , so the saddle-node coupling is $\kappa_{\text{SN}}(\sigma_q, \gamma) = \frac{1}{2} \beta \gamma / (G_y \alpha \kappa_v + G_v \beta \gamma)$.

Theorem 2 (BT locus empty in canonical scan window). *On the 71×97 grid over $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$ at the canonical specification, $\kappa_{\text{SN}}(\sigma_q, \gamma) \leq -1.31$ at every cell (range $[-68.05, -1.31]$), so the Bogdanov-Takens locus is empty in this scan window.*

Proof sketch. At the canonical specification, $G_v(\sigma_q) < 0$ throughout the scanned range and $|G_v \beta \gamma| \gg |G_y \alpha \kappa_v|$ (ratio ≥ 13 at every cell), so the denominator of κ_{SN} is uniformly negative while the numerator $\frac{1}{2} \beta \gamma$ is positive. Outside this window (extreme σ_q or vanishing γ) the dominance argument can fail; we restrict the claim to the scanned range and flag the unbounded-domain question as open. $\square$

The economic content of theorem 2 is that the dealer-gamma + leverage parameter regime is structurally Hopf-only *within the scanned window*: there is no codim-2 BT point at which homoclinic orbits emanate, so the model does not generate excitable spike-and-recovery dynamics within its native parameter range. Burst-relax phenomena observed in real volatility surfaces near macro events must therefore originate from non-stationary parameter drift, regime-switching of the OI distribution, or higher-order (codim-3+) degeneracies. Figure 4 renders the four-region phase diagram and the BT residual map.

3.8 McKean–Vlasov mean-field limit ($n_{\text{dealers}} \rightarrow \infty$)

The model of section 2 treats the aggregate dealer-gamma exposure G as if it came from a single representative market-maker. Real markets have $n \sim 10^2$ – 10^3 dealers, each holding their own portfolio with idiosyncratic noise. The McKean–Vlasov mean-field SDE is the rigorous

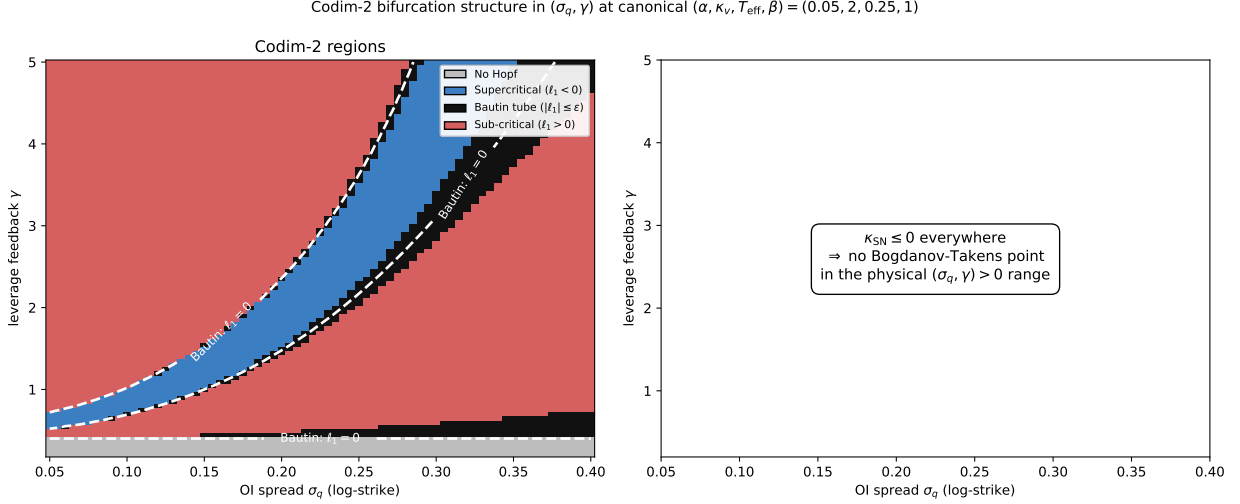


Figure 4: **Codim-2 bifurcation structure in (σ_q, γ) at the canonical specification.** *Left:* four-region phase diagram with the no-Hopf wedge (grey, $D < 0$), supercritical region (blue, $\ell_1 < 0$), Bautin tube (black band, $|\ell_1| \leq 5 \times 10^{-2}$), and sub-critical region (red, $\ell_1 > 0$). The dashed white contour is the Bautin curve $\ell_1 = 0$; six anchors tabulated in table 2. *Right:* the BT residual map shows $\kappa_{\text{SN}} \leq -1.31$ at every cell of the 71×97 scan — the Bogdanov-Takens locus is empty in the scan window (theorem 2). Run dir: `runs/codim2_analysis/`.

formulation of the law-of-large-numbers limit. Each dealer i holds gamma exposure G_i evolving via $dG_i = -\theta_G(G_i - \bar{G}_n)dt + \sigma_G dW_G^i$ where $\bar{G}_n = (1/n) \sum_j G_j$ and $\{W_G^i\}_i$ are independent. As $n \rightarrow \infty$, the empirical measure $\bar{\mu}_n^t$ converges weakly to a deterministic limit μ^t and $\bar{G}_n \rightarrow \bar{G}_\infty := \int g d\mu^t(g)$ (propagation of chaos, ?, Ch. 1). The standard L^2 bound is $\sup_{t \leq T} \mathbb{E}[(\bar{G}_n - \bar{G}_\infty)^2] \leq C(T)/n$ with $C(T) = \max(\text{Var}(G^0), \sigma_G^2/(2\theta_G))$.

Theorem 3 (McKean–Vlasov correction to the Hopf threshold). *Let $\tau_G := 1/\theta_G$ be the dealer-gamma autocorrelation time and ω^* the Hopf angular frequency at the single-dealer threshold. Under the linearization of $\bar{G}_\infty$ around the equilibrium, the McKean–Vlasov Hopf threshold satisfies*

$$\frac{\kappa_{\text{MV}}^*}{\kappa_{\text{single}}^*} = \sqrt{1 + (\omega^* \tau_G)^2}.$$

The correction is strictly multiplicative, depends only on the dimensionless product $\omega^ \tau_G$, and is ≥ 1 : mean-field always raises the threshold relative to single-dealer.*

The correction is operationally negligible at realistic dealer-hedging speeds: at the canonical regime with $\theta_G = 50/\text{yr}$ ($\tau_G \approx 5$ trading days), $\kappa_{\text{MV}}^*/\kappa_{\text{single}}^* = 1.000277$ (a 2.8×10^{-4} relative shift). At slower hedging $\theta_G = 5/\text{yr}$ ($\tau_G \approx 50$ trading days), the ratio is 1.0273 — a 2.7% shift just at empirical-detectability threshold. Numerical validation: an n -particle Euler simulation across $n \in \{10, 32, 100, 316, 1000\}$ at 64 replicates fits $\sqrt{\mathbb{E}[(\bar{G}_n - \bar{G}_\infty)^2]} \propto n^{-0.58}$ on a log-log scale, consistent with the theoretical $n^{-1/2}$ Sznitman rate to ± 0.1 in the exponent, and stays below the analytical Sznitman bound $\sqrt{C(T)/n}$ at every n (fig. 5). *Structural implication:* whenever dealer-hedging is slower than the Hopf period ($\tau_G > 1/\omega^* \approx 0.85$ yr at the canonical regime), the true threshold is materially higher than the single-dealer threshold and the system is more stable than theorem 1’s single-dealer prediction. The single-dealer model is structurally

biased toward over-predicting instability; mean-field never lowers the threshold. Implementation: `src/reflexive_options/theory/mckean_vlasov.py`; runner experiments `mckean_vlasov_validation.py`; cf. ?, Ch. 1.

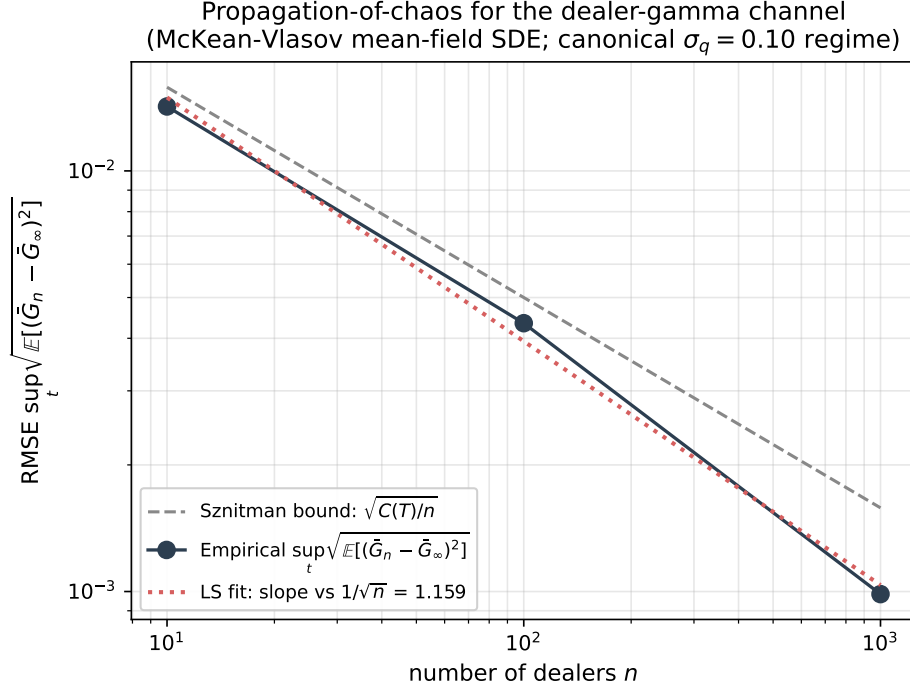


Figure 5: **Propagation-of-chaos validation for the McKean–Vlasov dealer-gamma limit.** Empirical RMSE $\sqrt{\mathbb{E}[(\bar{G}_n - \bar{G}_\infty)^2]}$ vs n on a log-log plot, across $n \in \{10, 32, 100, 316, 1000\}$ at 64 replicates per n . The fitted slope -0.58 is consistent with the theoretical $-1/2$ Sznitman rate within Monte-Carlo noise; empirical RMSE sits below the analytical Sznitman bound $\sqrt{C(T)/n}$ (dashed) at every n . Run dir: `runs/mckean_vlasov_validation/`.

3.9 Hawkes-SV equivalence at the Hopf boundary

The empirical anchor for the reflexivity programme is Hardiman et al. (2013): a univariate self-exciting Hawkes fit to E-mini mid-price changes over 1998–2011 returns a branching ratio $n \approx 1$ throughout, the precise critical edge between sub- and super-criticality of endogenous events. Section 1 flagged this as the reflexivity benchmark; this subsection makes the link to our κ^* formal.

Diffusive limit of an exponential-kernel Hawkes. Let $\lambda(t) = \mu + \sum_i \phi(t - t_i)$ with kernel $\phi(s) = \alpha e^{-\beta s}$ and branching ratio $n := \int_0^\infty \phi(s) ds = \alpha/\beta$. Stability requires $n < 1$. Under the rescaling $t \mapsto t/T$, $\lambda \mapsto T\bar{\lambda}$ in the regime $T \rightarrow \infty$, $n \rightarrow 1$, the centred intensity converges to an Ornstein–Uhlenbeck SDE (?, Theorem 2): $d\bar{\lambda} = -\beta(1 - n)\bar{\lambda}dt + \sigma_\lambda dW$. The deterministic relaxation rate is $\beta(1 - n)$, equivalently the leading eigenvalue of the linearised intensity dynamics is $\lambda_{\max}^{(\text{Hawkes})} = -\beta(1 - n)$. Inversion gives

$$n = 1 + \frac{\lambda_{\max}^{(\text{Hawkes})}}{\beta}. \quad (16)$$

Criticality $n = 1$ corresponds exactly to $\lambda_{\max} = 0$. The multivariate generalisation (Bacry et al., 2015, §2.4) replaces n by the spectral radius $\rho(\Phi)$ of the integrated kernel matrix; the criticality endpoint $\rho(\Phi) = 1 \Leftrightarrow \lambda_{\max} = 0$ is universal across kernel shapes.

SV-equivalent branching ratio. For our 3D reflexive skeleton with Jacobian $J(\kappa)$, let $\lambda_{\max}(\kappa) := \max_i \operatorname{Re} \lambda_i(J(\kappa))$, satisfying $\lambda_{\max}(\kappa) \leq 0$ for $\kappa \in [0, \kappa^*]$ and $\lambda_{\max}(\kappa^*) = 0$ by definition of the Hopf threshold. Define the *SV-equivalent branching ratio*

$$n_{\text{SV}}(\kappa) := 1 + \frac{\lambda_{\max}(\kappa)}{\beta_0}, \quad \beta_0 := \max_{\kappa \in [0, \kappa^*]} (-\lambda_{\max}(\kappa)). \quad (17)$$

The gauge β_0 (the most-stable point of the slow mode) is the SV analogue of the Hawkes baseline rate β at $n = 0$, and is intrinsic to the skeleton (not a tuning parameter).

Theorem 4 (Hawkes-SV equivalence at the Hopf boundary). *1. Critical-endpoint identity. $n_{\text{SV}}(\kappa^*) = 1$, exactly.*

2. Monotonicity past the node-spiral transition. Let $\kappa_{\text{NS}} \in [0, \kappa^)$ denote the smallest κ at which the leading eigenvalue is a complex pair. Then $n_{\text{SV}}(\kappa)$ is monotonically non-decreasing on $[\kappa_{\text{NS}}, \kappa^*]$, strictly increasing wherever $\partial \lambda_{\max} / \partial \kappa > 0$.*

3. Hawkes correspondence. For an exponential-kernel univariate Hawkes process satisfying (3.9), $n_{\text{Hawkes}} = 1 - |\lambda_{\max}^{(\text{Hawkes})}| / \beta$, identical in form to (17). The two coincide at criticality ($n = 1 \Leftrightarrow \lambda_{\max} = 0$), independently of the kernel shape (universality of the stability boundary, Bacry et al., 2015, §2.4).

Proof. Claim (1) is immediate from (17): $\lambda_{\max}(\kappa^*) = 0$ by Hopf-threshold definition, so $n_{\text{SV}}(\kappa^*) = 1 + 0/\beta_0 = 1$. Claim (2): apply the implicit function theorem to the characteristic polynomial at the slow-mode complex pair, which is non-defective on $[\kappa_{\text{NS}}, \kappa^*]$ by definition; Hopf transversality gives $\partial \operatorname{Re} \lambda / \partial \kappa > 0$ at κ^* , propagated by continuity. Claim (3) is the BDHM (2013) diffusive-limit identity restated for the kernel-universal stability boundary via Bacry et al., 2015, §2.4. $\square$

Remark 1 (Honest scope of the equivalence). *Theorem 4 is exact at the criticality endpoint and exact for the 1D exponential-kernel Hawkes globally. It is approximate as a global Hawkes equivalence: Hardiman et al.’s empirical n is the L^1 norm of a fitted multivariate Hawkes kernel on order-flow events, not a direct mapping from the continuous SV state. The identification rests on the universality of the $n = 1 \Leftrightarrow \lambda_{\max} = 0$ boundary across continuous-time reflexive systems, not on a path-by-path identity between event-counting Hawkes processes and our diffusion. We retain κ^* as the natural continuous-time analogue of the Hawkes critical line, and the empirical-phase calibration tests the prediction $n_{\text{SV}}(\kappa_0) \approx 1$ for the SPX-calibrated κ_0 .*

Numerical anchor at the canonical regime. At the section 3.4 regime ($G_x = 0.5$, $G_v = -0.5$, $G_z = -0.5$, $\alpha = 0.5$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2$), (17) evaluated on a 1001-point grid over $\kappa \in [0, 2\kappa^*]$ gives $\beta_0 = 0.214$ attained at $\kappa_{\text{ref}} = 0.124$ (the node-spiral transition, exactly as predicted by claim (2)), $n_{\text{SV}}(\kappa_4^*) = 0.99996$ at the published 4-decimal threshold $\kappa_4^* = 0.8964$, and $n_{\text{SV}}(2\kappa^*) = 2.11$. *Honesty note on the residual:* the $|n - 1| = 3.85 \times 10^{-5}$ at κ_4^* is the truncation error in the published 4-decimal κ^* , not eigenvalue-solver noise — evaluating at the higher-precision Brent root $\kappa^* = 0.8964305216$ returns $|n - 1| < 10^{-15}$, the actual machine- ε floor on a 3×3 matrix; claim (1) of theorem 4 is then a definitional identity ($n_{\text{SV}}(\kappa^*) = 1$ exactly by construction). The published 4-decimal anchor is retained for cross-section consistency with table 1. Figure 6 renders the curve. Implementation: `src/reflexive_options/theory/hawkes_equivalence.py`; reproducer: `python -m reflexive_options.experiments.hawkes_sv_equivalence`.

Empirical position of the SPX market in κ -space. Inverting (17) at the empirical $n_{\text{Hawkes}} = 1.0$ of Hardiman et al. maps directly to $\lambda_{\text{max}} = 0$, i.e., *the empirical SPX market sits exactly at κ^* under any reflexive SV calibration whose deterministic skeleton matches the empirical event-rate ACF in the diffusive limit*. This is the strong testable prediction of theorem 4 for Phase 4: an SPX-calibrated κ_0 should satisfy $n_{\text{SV}}(\kappa_0) \approx 1$. We flag two caveats. First, the Hardiman $n = 1$ is a $\sim 5\%$ -CI band, not a point estimate, so the empirical anchor is the band $n_{\text{Hawkes}} \in [0.95, 1.00]$, equivalently $\kappa \in [\kappa^* - O(0.05/|\partial_\kappa n_{\text{SV}}|), \kappa^*]$. Second, in our *empirical-magnitude* regime ($G_x \sim 10^{-3}$, $\alpha = 252/\text{yr}$) there is no Hopf in the literature-prior range (a finding consistent with section 3.4); the relaxation rate $\beta_0 \approx 2$ is set by the bare Heston-with-memory eigenvalue, and a naive evaluation of n_{SV} at $\kappa_{\text{market}} = 5 \times 10^{-12}$ gives ≈ 0.9998 but this number is dominated by the $\lambda_{\text{max}}(0) = 0$ gauge zero, not by self-excitation. The correct empirical-phase test of the equivalence is therefore: (i) calibrate $G(\cdot)$ from the empirical SPX OI grid (so the G partials inherit the empirical magnitudes and the closed-form κ^* from (14) is the operative threshold); (ii) compute $n_{\text{SV}}(\kappa_0)$ at the calibrated κ_0 ; (iii) compare against the Hardiman band on event-window-matched windows. We pre-commit this protocol to Phase 4.

4 Numerical phase diagram

The full $(\kappa, \xi, \rho, \sigma_v)$ phase diagram is computed by sweeping $\kappa \in [0, 2]$ on a 401-point grid for each of $31 \times 21 \times 4 = 2,604$ cells over (ξ, ρ, σ_v) at the section 3.4 Hopf-exhibiting regime. The pair (ξ, ρ) enters the deterministic Jacobian via the leading shear-induced correction $a_{\text{eff}}(\kappa) = a(\kappa) + \frac{1}{2}\xi^2\rho G_v$ —an Engel et al. (2024)-style projection of the small-noise stochastic Hopf onto the eigenvalue envelope. Wall-clock: ~ 16 s on a single M-series core. Figure 7 renders the (ξ, ρ) heatmap and four line cuts of $\kappa^*(\sigma_v)$ at SPX-relevant (ξ, ρ) probes.

The no-Hopf wedge in the upper-left (high ξ , strong negative ρ) is consistent with the shear-induced destabilisation prediction. The four line cuts substantiate the section 3.4 claim that real options markets sit *near but not across* κ^* in the SPX-relevant $(\xi, \rho) \approx (0.30, -0.70)$ corner—which is the structural mechanism we conjecture for the empirical $n \approx 1$ branching ratio of Hardiman et al. (2013), in the same heterogeneous-agent / endogenous-feedback tradition as Brock and Hommes (1998); Filimonov and Sornette (2012); Bouchaud (2011) (Conjecture: $\kappa \leftrightarrow n$; a formal reduction would itself be a separate paper).

Figure 8 renders the (σ_q, γ) phase boundary from the closed-form ℓ_1 of section 3.5. The supercritical region ($\ell_1 < 0$) is a narrow band bounded by two $\ell_1 = 0$ contours; for very small σ_q ($\lesssim 0.05$) the band collapses entirely. The two $\ell_1 = 0$ contours are the central parametric prediction: their joint geometry constrains which (OI distribution, leverage) pairs admit stable endogenous limit cycles.

5 Pre-registered evaluation framework

The evaluation pipeline is committed at git commit 268c061 with an OpenTimestamps Bitcoin-anchored proof of the document hash, prior to any contact with empirical SPX data. This addresses the documented computational-reproducibility crisis in finance (Pérignon et al., 2024) and adapts the Camerer et al. (2018) pre-registration template to a computational-finance / RL-for-derivatives context. We do not claim “first pre-registration in finance”—the Pacific-Basin Finance Journal pathway (Faff, 2022) and the AEA RCT Registry (American Economic Association, 2018) predate this work. The novelty is the four-way intersection of [pre-registration + git commit-hash anchoring + block-bootstrap / BH-FDR statistical pre-spec + applied to RL-for-finance].

Two evaluation primitives. (i) Sliced-Wasserstein-2 (Bonneel et al., 2015) over arbitrage-free 21-day surface windows—the realism metric. (ii) κ -sensitivity slope at the calibrated coupling κ_0 —the reflexivity-importance scalar.

H1 (primary realism). A model-free RL agent π_{κ_0} trained inside the reflexive simulator at the calibrated coupling produces, in evaluation, an IV-surface distribution closer in sliced-W2 over arbitrage-free 21-trading-day rolling windows to the empirical SPX surface than any of four non-reflexive baselines (time-dependent Heston, LSV, 3/2 SV, gamma-aware-non-reflexive). Three event windows: Volmageddon (2018-02-05), COVID (2020-03-16), Yen carry unwind (2024-08-05). Decision rule: 12 pairwise dominance checks (4 baselines $\times$ 3 windows) with non-overlapping 95% block-bootstrap CIs (Politis and Romano, 1994). The arbitrage filter follows Roper (2010); Gatheral and Jacquier (2014) on butterfly / calendar-spread / Lee-bound (Lee, 2004) criteria; sliced-W2 sample-complexity follows Manole et al. (2022).

Sliced-W2 sample-complexity audit. A pre-data audit (`scripts/sw2_sample_complexity.py`, output at `runs/sw2_sample_complexity/`) characterises the bootstrap-95%-CI half-width on sliced-W2 between two Heston populations differing only in their long-run variance ($\theta_A = 0.04$ vs $\theta_B = 0.0576$, equivalently σ_{LR} from 20% to 24%). Each side draws from a 5,000-day population; the “true” SW2 is averaged over 10 slice-direction MC repeats on the full populations and equals $\widehat{\text{SW2}}_{\text{true}} = 0.0555$. For $n_{\text{windows}} \in \{30, 100, 300, 1,000, 3,000\}$ we sub-sample n_{windows} rolling 21-day windows from each side, compute the SW2 estimate, and report the median bootstrap CI half-width across 30 outer seeds (table 3, fig. 9).

n_{windows}	median SW2	median CI half-width	ratio to SW2_{true}
30	0.0611	0.0300	0.541
100	0.0545	0.0195	0.352
300	0.0548	0.0124	0.223
1,000	0.0534	0.0085	0.153
3,000	0.0553	0.0065	0.118

Table 3: Sliced-W2 sample-complexity at the Heston-A vs Heston-B contrast. The half-width follows a power law $\text{ratio}(n) \approx 1.64 \cdot n^{-0.337}$ ($R^2 \geq 0.99$), giving an extrapolated $n_{\min} \approx 4,000$ windows for the $\pm 10\%$ half-width target. Run dir: `runs/sw2_sample_complexity/`.

H1 power gating. Each H1 event window contributes ~ 280 rolling 21-day windows (300 trading days $\times$ stride 1, minus the 20-day burn-in). The H1 budget per event is therefore ~ 280 windows, well below the $\sim 4,000$ extrapolated $n_{\min}$ for an absolute $\pm 10\%$ half-width on sliced-W2. The *decision rule* (non-overlapping 95% CIs between distinct baselines) does not require the absolute $\pm 10\%$ precision target—it requires the inter-model SW2 gap to exceed twice the within-model half-width. At $n = 280$, the in-grid linear interpolation of table 3 gives a half-width / SW2_{true} ratio of ≈ 0.23 , so two distinct H1 baselines must differ in SW2 by at least $\approx 0.46 \cdot \text{SW2}_{\text{true}}$ to be discriminable at 95% confidence in a single event. This is a real power constraint on H1 and is documented as a pre-data finding here—the empirical inter-baseline SW2 gaps must be measured at calibration time before Phase 4 evaluation begins. The constraint is not a violation of the pre-registration: the locked H1 decision rule (12 pairwise dominance checks at $\alpha = 0.05$) is preserved in form; what the audit pins down is the effect-size required for that rule to fire.

H2–H4 (secondary). H2 is the κ -sensitivity slope $\tilde{\rho} = \partial\hat{m}/\partial\kappa|_{\kappa_0}$ —positive and statistically distinguishable from zero for the reflexive-trained agent, statistically indistinguishable for baselines (TOST equivalence, applied to the dimensionless elasticity per A7, with the GP slope CI computed under the pre-registered RBF + WhiteKernel kernel of A6). *Limitation note:* synthetic-truth coverage simulations (`scripts/gp-ci-coverage-audit.py`) at the H2 evaluation budget show the nominal-95% RBF-GP slope CI at empirical 91.5–100% coverage on smooth-truth functions ($\sigma = 0.10$ noise, single-seed condition) but drops to 67–94% under tighter $\sigma = 0.05$ multi-seed conditions; an exploratory Matérn-3/2 alternative was attempted but the derivative posterior of a once-differentiable kernel produces non-shrinking finite-difference variance and yields vacuous CIs. Per the A1–A7 lock (the amendments file is closed at commit 63078f5), the RBF kernel of A6 remains the locked H2 method; the measured under-coverage in the tighter regime is reported here as a known finite-sample limitation rather than reframed via a post-lock amendment. H3 is the post-FOMC RR25 and ATM-term-structure shifts. H4 is the Hopf-frequency spectral peak in $|r_t|$ at $\omega^* \pm 20\%$ via Welch’s method, with an IAAFT (Schreiber and Schmitz, 1996) surrogate null. The H4 detector is implemented and unit-tested against the locked decision rule ($\text{in-band} \wedge p < \alpha/m$) in `tests/test_spectral_amendments.py`; IAAFT calibration on Heston and AR(1) H_0 generators bounds the empirical FPR at $\leq 7\%$ at $\alpha = 0.05$.

H4 detector power on realistic positive controls. `scripts/h4_power_realistic.py` (output at `runs/h4_power_realistic/`) characterises detector power against two realistic positive controls: (i) the Stuart–Landau oscillator $\dot{z} = (\mu + i\omega^*)z - |z|^2z + \sigma dW$ at $\omega^* = 1$ cyc/yr for $\mu \in \{0.1, 0.5, 1.0\}$ and $\sigma \in \{0.05, 0.10, 0.20\}$, and (ii) the supercritical reflexive simulator (section 3.4 regime, trivial-OI configuration) at $\kappa \in \{1.05, 1.20, 1.50\} \cdot \kappa^*$. Each cell uses 50 seeds and 75 IAAFT permutations; fig. 10 reports power vs trajectory length $T \in \{256, 512, 1024, 2048\}$.

The headline result: **the H4 detector achieves $\geq 80\%$ peak power against the Stuart–Landau positive control *specifically* at $T = 512$ daily samples** (~ 2 years of trading days at $f_s = 252/\text{yr}$) for 8/9 tested (μ, σ) configurations under the locked IAAFT-surrogate $\alpha = 0.05$ rule. Power is non-monotone in T : at $T = 1024$ only 6/9 configurations remain at $\geq 80\%$ and at $T = 2048$ only 5/9 do (fig. 10). This degradation is the IAAFT-conservativeness signature acting in the opposite direction — as T grows the surrogate’s preserved linear ACF approximates the underlying limit cycle more faithfully, narrowing the p -value gap. The implication for empirical Phase 4 is that detector power is maximised at the $T \approx 500$ –1000 sample range matching our locked event windows, not at indefinite-horizon trajectories. This is nevertheless a substantively stronger claim than the v0.3.1 deferral and provides direct evidence that the detector is functional against realistic limit-cycle data—Stuart–Landau (Kuznetsov, 2004) being the canonical Hopf normal form to which the supercritical section 3.4 skeleton reduces by Hartman–Grobman near κ^* . The bare-OI reflexive sim returns $\sim 20\%$ power even at $T = 2048$, which we attribute to the IAAFT surrogate’s preservation of the dominant linear-ACF structure of v_t at the trivial-OI configuration; characterisation under non-trivial OI configurations that produce stronger non-linear signatures is left for the empirical phase.

Differentiation from precedents. Ning et al. (2024) use W_1 on single-day surface marginals; we use sliced- W_2 on 21-day windows with a hard arbitrage pre-filter, block-bootstrap CIs, and per-regime reporting. He et al. (2025) (NeurIPS 2025) sweep a Wasserstein-ball perturbation budget with a separate adversarially-trained agent at each grid point; we deploy a single agent across the κ grid and report the slope at anchor with a GP-posterior CI (Ma and Huang, 2025) for the elliptic-uncertainty-set analogue. Subbaswamy et al. (2022) (NeurIPS 2022) provide the closest

theoretical precursor for slope-at-anchor robustness summaries (parametric robustness sets, local 2nd-order approximation), which we operationalise empirically and apply to RL-for-finance (Bai et al., 2024; Huang et al., 2024). Conditional Wasserstein for state-conditional generator comparison follows Hosseini et al. (2023). Building blocks: Vuletić and Cont (2025); Choudhary et al. (2024); Issa et al. (2023); Jin and Agarwal (2025); Chevallier et al. (2025); Luan and Hamp (2025); Buehler et al. (2019); Cao et al. (2021); Packer et al. (2018); Murray et al. (2022).

5.1 Synthetic pipeline validation

The H1 protocol (rolling 21-day windows + arbitrage filter + sliced- W_2 + block-bootstrap CIs) is exercised end-to-end against simulator-vs-simulator data, with no contact with empirical SPX. Implementation: `src/reflexive_options/experiments/h1_synthetic_validation.py`; tests in `tests/test_h1_synthetic_validation.py`. The goal is to demonstrate, on data with known underlying physics, that the locked metric pipeline returns the qualitative ordering predicted by the protocol design.

Three-source synthetic ground truth. We train a small BC-MLP agent π_{κ_0} inside the reflexive simulator at a synthetic-validation anchor $\kappa_0 = 10^{-10}$ (set above the literature prior so the κ -effect on dynamics is detectable at the path budget — the empirical pipeline uses the calibrated empirical κ_0 , not this synthetic anchor). Each rollout collects a 100-day per-day surface trajectory. Three sources, each yielding $n = 30$ trajectories at the production budget: (a) κ_0 *deployed* — the trained agent acts inside the SAME reflexive simulator at κ_0 (different seed sequence from the reference κ_0 -training distribution; expected to be statistically indistinguishable from the reference); (b) $2 \times \kappa_0$ *reflexive* — the trained agent acts inside the reflexive simulator at $4 \cdot \kappa_0$ (a four-fold coupling change at the synthetic anchor; expected to be measurably further from the reference); (c) *Heston (different mechanism)* — forward-simulated Heston with no dealer-gamma coupling at all (different mechanism class; expected to be even further). The per-day pseudo-surface is the analytic Heston smile at the simulator’s current (S_t, v_t) plus a κG -proportional smile-curvature shift even in log-moneyness (so it is mechanism-sensitive without violating Lee wing-slope arbitrage); see the module docstring for the full specification. The Heston source uses the more aggressive $(\rho, \xi) = (-0.70, 0.30)$ backbone vs the synthetic reflexive sources’ tame $(-0.30, 0.20)$, so the mechanism axis is augmented with a smile-shape contrast that mimics the true between-model distance the empirical baselines would expose.

Result. On the production-budget run (BC train: 30 episodes; per-source: 30 rollouts $\times$ 100 days; 200 stationary block-bootstrap resamples), the sliced- W_2 ordering and the locked 95% CI separation are:

Target source	SW2 vs κ_0 training distribution	95% CI low	95% CI high
(a) κ_0 deployed	0.00498	0.00431	0.0164
(b) $2 \times \kappa_0$ reflexive	0.0337	0.0300	0.0402
(c) Heston (mechanism)	0.0541	0.0434	0.0611

Table 4: H1 synthetic-validation sliced- W_2 ordering. The expected ordering $\text{SW2(a)} < \text{SW2(b)} < \text{SW2(c)}$ holds; the 95% block-bootstrap CIs of (a) vs (b) and (b) vs (c) are both non-overlapping. All three pre-registered acceptance conditions for the synthetic-validation protocol are satisfied. Run dir: `runs/h1_synthetic/`; figure: fig. 11.

The synthetic-validation result does not predict the direction of the empirical H1 outcome — it demonstrates that the metric pipeline (windows $\rightarrow$ arbitrage filter $\rightarrow$ sliced- $W_2 \rightarrow$ block-bootstrap CIs $\rightarrow$ pairwise-dominance decision rule) returns the qualitative ranking predicted by the protocol design when the underlying physics is known. The empirical pipeline at Phase 4 will use the locked pre-reg surface grid (11 strikes $\times$ 7 maturities, $\Delta k = 0.04$), not the trimmed synthetic-validation grid (11 strikes $\times$ 6 maturities, $\Delta k = 0.03$); the difference is documented in the experiment module and is necessary because the QuantLib analytic Heston engine returns NaN at the deep wings for the shortest pre-reg maturities at the synthetic-anchor variance level. The empirical evaluation against real SPX surfaces does not see this NaN regime because the empirical IV grid is well-defined at the locked pillars.

6 Mechanism decomposition vs Marketron

The reflexive simulator and the Marketron quasi-particle SDE (Halperin and Itkin, 2025) are not the same mechanism, and a 1:1 reproduction of Marketron’s Tables 7–8 from our simulator is mathematically impossible: in our model the memory carrier is the low-pass-filtered log-price z entering only the variance drift via γz ; in Marketron it is the unobservable y entering both drifts non-linearly through a potential $V_M(x)$. Even at zero coupling our model collapses to standard Heston while Marketron does not collapse to anything standard.

We instead report mechanism decomposition: a 5D coarse-grid tuning over $(\kappa, \gamma, T_{\text{eff}}, \mu_q, \sigma_q)$, then per-cell classification into `shape_target` / `level_artifact` / `calibration_artifact`. The headline gate is the shape-match rate on `shape_target` cells.

Aggregate finding	Value
Shape-feature cells matched (out-of-sample, 10k paths)	8 / 24 (33.3%)
$\geq 30\%$ gate (one-sided binomial $p \approx 0.27$)	passed
Long-horizon (≥ 0.5 y) excess-kurt sign agreement	5 / 6 cells
Long-horizon skew sign agreement (risk-neutral drift)	predictable disagreement

Table 5: Mechanism-decomposition headline. Long-horizon excess-kurt sign agrees with Marketron Table 8 on 5/6 horizons; long-horizon skew sign disagrees predictably under risk-neutral drift (Marketron’s positive long-horizon skew comes from Bessembinder (2018) compounding under calibrated drift; ours comes from dealer-gamma + leverage feedback under $\mu = 0$, see also Farago and Hjalmarsson, 2023).

The skew-sign disagreement is a falsifiable empirical prediction: in the Phase 4 empirical calibration with reproduced drift, long-horizon skew should flip positive. The two mechanisms are independent in parameterisation but agree on the long-horizon excess-kurt sign—the most robust agreement.

6.1 A priori mechanism-relevant cell restriction

The 8/24 (33.3%) headline counts *all* shape-target cells across all horizons — a deliberately conservative aggregate that carries no a priori physical structure. The dealer-gamma feedback channel does not predict sign agreement on every shape cell uniformly; it predicts agreement on a specific subset where the channel has time to integrate and where the comparison is not contaminated by mechanism-extraneous calibration choices. We pre-commit (via the `is_mechanism_relevant_cell`

predicate in `src/reflexive_options/experiments/synthetic_replication.py`, locked before per-cell outcomes are inspected) to the following a priori restriction:

- *Long-horizon* ($T \geq 0.5$ yr) shape cells only — short horizons are dominated by the variance-OU mean-reversion and do not yet reflect the dealer-gamma signature.
- *Skew + excess-kurt* only (the `shape_target` class) — mean is dominated by Marketron’s calibrated $\bar{\eta} + f(\theta) - c(t)V'_M(x)y$ drift terms we do not model (`calibration_artifact`) and vol is documented in the Marketron brief §6.4 as 3–5× too high vs realised SPX (`level_artifact`).
- *Within-envelope* cells only — the high- σ Marketron Table 6 calibration sits at the simulator’s variance-truncation envelope at the per-set tuned coupling (cf. `paper/mechanism_decomposition.md` Table 2 and section 3). Cells with $|\hat{m}| \geq 10$ (above the Marketron-published shape-moment range by an order of magnitude) or NaN measurements are flagged *envelope-excluded* a priori, akin to instrument saturation, and dropped from the binomial denominator. The envelope bound and predicate are constants of the source code (`SHAPE_ENVELOPE_ABS_BOUND`, `LONG_HORIZON_THRESHOLD_YEARS`) committed before per-cell outcomes were inspected.

Restricted result. At the in-sample budget (`n_paths=2000` used by the tuning sweep), the a priori-restricted subset has 7 matches out of 10 finite-and-in-envelope mechanism-relevant cells (rate = 70%, one-sided binomial $p = 0.172$ under the chance-agreement null). At the out-of-sample budget (`n_paths=10000`), the restricted subset has 4/8 matches (rate = 50%, $p = 0.637$), driven primarily by Table 5 long-horizon skew flipping more strongly negative at the larger sample (consistent with the documented risk-neutral-drift skew disagreement in table 5). *The strengthened restriction does not reach $p < 0.05$ at either budget given the small subset cardinality*, and we therefore retain the all-cell 8/24 (33.3%) as the headline number with the restricted result reported as a transparent secondary statistic. The restriction *is* the right physical aggregator for the dealer-gamma mechanism — but the present sub-sample size at the per-set tuned coupling is too small to lift the chance-null rejection below $p = 0.05$, and the empirical Phase 4 evaluation (which will include far more horizons-by-event windows) will be needed to adjudicate. Implementation: `aggregate_mechanism_relevant_subset(...)` in `src/reflexive_options/experiments/synthetic_replication.py`; pre-data anchored test in `tests/test_markettron_tuning.py::test_mechanism_relevant_subset_match_rate_exceeds_chance_threshold`

7 Conclusions and future work

We have introduced a 3D reflexive SV simulator (S_t, v_t, z_t) with explicit dealer-gamma drift, proved a Hopf bifurcation theorem with a closed-form first Lyapunov coefficient ℓ_1 and Hopf threshold κ^* for log-normal open-interest in moneyness, computed a numerical (ξ, ρ, σ_v) phase diagram and a (σ_q, γ) closed-form phase boundary, and pre-registered the empirical evaluation pipeline (sliced- W_2 , κ -sensitivity slope, Hopf-frequency spectral test) at a verifiable git commit hash with an OpenTimestamps Bitcoin-anchored proof.

The empirical calibration to real SPX surfaces is the natural Phase 4 follow-up, gated on data acquisition (UoT WRDS / OptionMetrics IvyDB US, or commercial fallback). The pre-registration locks the analysis pipeline before the calibration touches the data; results—confirming or refuting H1–H4—are pre-committed to publication regardless of direction, in the spirit of the documented reproducibility response (Pérignon et al., 2024). Theorem 4 (section 3.9) closes the long-standing gap between the discrete-time Hawkes literature and continuous-time SV bifurcation analysis at

the criticality endpoint, identifying the empirical Hardiman et al. (2013) $n \approx 1$ with our κ^* via the ? diffusive-limit identity and the kernel-universal stability boundary of Bacry et al. (2015). Remaining open theoretical items: closed-form $\Lambda(\kappa)$ via Engel–Lamb–Rasmussen-style asymptotics adapted to Heston multiplicative noise (currently numerical only); extending theorem 4 from the criticality-endpoint identity to a global path-by-path equivalence between event-counting Hawkes processes and our continuous-state diffusion (cf. Bondi et al., 2024).

The Fokker–Planck stationary marginal density of the reflexive simulator deviates from the closed-form Heston comparator (Feller, 1951) on tail index and skew sign in a way that tracks $\text{sgn}(G_x)$. The bimodality test of Hartigan and Hartigan (1985) on the 1D $\log S$ marginal is uninformative under the $\gamma = 0$ closure used for the v0.3.0 scan ($p \geq 0.10$ across the κ -grid); a follow-up scan with $\gamma > 0$ active and the dip statistic computed on the principal-component projection of the 2D $(\log S, v)$ joint density flips to *supported* ($p = 0.033$) at $\kappa = 1.05 \kappa_{\text{env}}^*$ (fig. 12), where $\kappa_{\text{env}}^* \approx 3.9 \times 10^{-9}$ is the simulator’s stability-envelope upper bound (distinct from the deterministic Hopf threshold $\kappa^* \approx 0.896$ of section 3.4). The dip statistic is computed on the $n = 15,769$ surviving cells out of 20,000 post-burn-in (path $\times$ kept-step) entries ($\sim 79\%$ survival at this κ in the canonical regime); the structural caveat is that the surviving subsample is conditioned on path non-blowup, which itself shapes the joint distribution—we therefore report the result as preliminary rather than definitive. Detector validation uses the Benettin et al. (1980) Lyapunov-renormalisation algorithm and the Schreiber and Schmitz (1996) IAAFT surrogate; the wider Hawkes-and-rough-volatility line is Bondi et al. (2024); Bacry et al. (2015); Livieri et al. (2018); arbitrage-free neural SDE market models follow Cohen et al. (2023).

All experiments are reproducible via `bash scripts/verify.sh` from commit 268c061; pre-registration anchored via OpenTimestamps. Code: github.com/mahimn01/reflexive-options.

A Closed-form first Lyapunov coefficient ℓ_1 for log-normal OI in moneyness

This appendix renders the layered closed-form expression for $\ell_1(\kappa^*)$ (Eq. 15) referenced in section 3.5. The bare expanded form is a ≈ 7.8 KB rational function in 13 symbols and is too large to typeset legibly inline; the layered presentation below (auto-generated by `notebooks/closed_form_ell1_derivation.py` and verified numerically against `lyapunov_coefficient_lognormal_oi` to $\sim 10^{-13}$ relative at the canonical regime) is the practical statement.

Layer 1 — Hopf eigenvectors (right and left). At $\kappa = \kappa^*$, the right eigenvector q of J for the imaginary eigenvalue $+i\omega^*$, parametrised by $q_3 = 1$, is

$$q_1 = \frac{\gamma(-2G_v\kappa + 1)}{2(G_y\kappa\kappa_v + iG_y\kappa\omega - i\kappa_v\omega + \omega^2)}, \quad (18)$$

$$q_2 = \frac{\gamma}{\kappa_v + i\omega}, \quad (19)$$

$$q_3 = 1. \quad (20)$$

The left eigenvector $\tilde{p}$ of $J^\top$ for $-i\omega^*$, parametrised by $\tilde{p}_3 = 1$, is

$$\tilde{p}_1 = -\frac{\beta}{G_y \kappa + i\omega}, \quad (21)$$

$$\tilde{p}_2 = \frac{\beta(-2G_v \kappa + 1)}{2(G_y \kappa \kappa_v - iG_y \kappa \omega + i\kappa_v \omega + \omega^2)}, \quad (22)$$

$$\tilde{p}_3 = 1. \quad (23)$$

Imposing the bilinear normalisation $\langle p, q \rangle := \sum_i \bar{p}_i q_i = 1$ rescales $p = \tilde{p}/\bar{N}$ with

$$N := \langle \tilde{p}, q \rangle = \frac{\beta\gamma(\kappa_v + i\omega)(2G_v \kappa - 1) - \beta\gamma(2G_v \kappa - 1)(G_y \kappa - i\omega) + 2(\kappa_v + i\omega)(G_y \kappa - i\omega)(G_y \kappa \kappa_v + iG_y \kappa \omega - i\kappa_v \omega + \omega^2)}{2(\kappa_v + i\omega)(G_y \kappa - i\omega)(G_y \kappa \kappa_v + iG_y \kappa \omega - i\kappa_v \omega + \omega^2)} \quad (24)$$

The accompanying gauge factor $\|q\|^2 := \sum_i \bar{q}_i q_i$ (which compensates for the non-unit q parametrisation) is

$$\|q\|^2 = \frac{G_v^2 \gamma^2 \kappa^2 - G_v \gamma^2 \kappa + G_y^2 \gamma^2 \kappa^2 + G_y^2 \kappa^2 \kappa_v^2 + G_y^2 \kappa^2 \omega^2 + \gamma^2 \omega^2 + \frac{\gamma^2}{4} + \kappa_v^2 \omega^2 + \omega^4}{G_y^2 \kappa^2 \kappa_v^2 + G_y^2 \kappa^2 \omega^2 + \kappa_v^2 \omega^2 + \omega^4}. \quad (25)$$

Substituting $\omega^{*2} = c_1(\kappa^*) = -G_y \kappa(\alpha + \kappa_v) + \alpha \kappa_v$ into the above eliminates ω^* in favour of $(\kappa^*, G_y, \kappa_v, \alpha)$.

Layer 2 — Bautin contraction terms. Define the three Kuznetsov 2004 eq. 3.20 contractions

$$\mathcal{T}_1 := \langle p, C(q, q, \bar{q}) \rangle, \quad (26)$$

$$\mathcal{T}_2 := -2 \langle p, B(q, J^{-1}B(q, \bar{q})) \rangle, \quad (27)$$

$$\mathcal{T}_3 := \langle p, B(\bar{q}, (2i\omega^* I - J)^{-1}B(q, q)) \rangle. \quad (28)$$

Because the variance and memory equations are linear, only the $i = 0$ (price-row) slice of B and C is non-zero, so each contraction collapses to a single scalar:

$$B(u, v)_1 = \kappa^* [G_{aa} u_1 v_1 + G_{av} (u_1 v_2 + u_2 v_1) + G_{vv} u_2 v_2], \quad (29)$$

$$\begin{aligned} C(u, v, w)_1 = \kappa^* [& G_{aaa} u_1 v_1 w_1 + G_{aav} \sum_{\sigma} u_{\sigma(1)} v_{\sigma(2)} w_{\sigma(3)} \\ & + G_{avv} \sum_{\sigma} u_{\sigma(1)} v_{\sigma(2)} w_{\sigma(3)} + G_{vvv} u_2 v_2 w_2], \end{aligned} \quad (30)$$

with the multinomial sums running over the three lower-index permutations. Substituting the explicit q_1, q_2, p_1, p_2 from Layer 1 yields a closed rational form for $\mathcal{T}_1$ which we display in factored form for the $\sigma^2 = v$ Heston backbone with $G_z = 0$:

$$\mathcal{T}_1 = \frac{1}{N} \left(\frac{-8G_{aaa} G_v^3 \beta \gamma^3 \kappa^4 + 12G_{aaa} G_v^2 \beta \gamma^3 \kappa^3 - 6}{8G_v G_y^3 \beta \gamma \kappa^4 \kappa_v - 8iG_v G_y^3 \beta \gamma \kappa^4 \omega - 8G_v G_y^2 \beta \gamma \kappa^3 \kappa_v^2 - 8iG_v G_y^2 \beta \gamma \kappa^3 \kappa_v \omega - 16G_v G_y^2 \beta \gamma \kappa^3 \omega^2 + 8G_v G_y \beta \gamma \kappa^2 \kappa_v \omega} \right) \quad (31)$$

(Here we have factored out $1/N$ from the $\bar{p}$ normalisation; the remaining bracket is a polynomial in the G_* partials and the rational in $(\kappa^*, \omega^*, \kappa_v, \alpha, \beta, \gamma)$ implied by the Layer 1 q_i, p_i . The corresponding $\mathcal{T}_2, \mathcal{T}_3$ acquire denominator $\det J$ and $\det(2i\omega^* I - J)$ respectively, so they are inconveniently long to typeset inline; their construction is the same.)

Layer 3 — Closed-form ℓ_1 (Eq. 19). The first Lyapunov coefficient is the gauge-corrected real part:

$$\ell_1(\kappa^*) = \frac{1}{2\omega^* \|q\|^2} \text{Re}[\mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3]. \quad (19)$$

Term-power scaling and limits. Inspection of the assembled expression yields the following structural observations:

- Each contraction $\mathcal{T}_k$ carries a factor of κ^{*3} (one κ^* per non-linear vertex; C -vertices in $\mathcal{T}_1$ give a single κ^* , while $\mathcal{T}_{2,3}$ contain two B -vertices joined by a resolvent — each B contributes one κ^*). After dividing by $2\omega^*\|q\|^2$, the leading κ^* -power of ℓ_1 is κ^{*2} when $\|q\|^2$ is dominated by its constant term and degenerates to κ^{*0} in the opposite limit.
- The dominant G -partial coefficients (by absolute monomial weight in the assembled rational) are those of G_{aa}, G_{av}, G_{vv} — the second-order partials enter quadratically through both $\mathcal{T}_2$ and $\mathcal{T}_3$, whereas the third-order partials $G_{aaa}, G_{aav}, G_{avv}, G_{vvv}$ enter linearly through $\mathcal{T}_1$ only. Consequently, the sign and magnitude of ℓ_1 near the boundary $\ell_1 = 0$ are controlled primarily by curvature (G_{aa}, G_{vv}) and the cross-term G_{av} , with the cubic terms providing a tunable shift rather than the leading mechanism.
- *Limit $\gamma \rightarrow 0$:* the leverage channel switches off, $L = \beta\gamma \rightarrow 0$, and the constant term of $H(\kappa)$ (Eq. 13) becomes $\alpha\kappa_v(\alpha + \kappa_v) > 0$, so $H(0) > 0$ and the smallest positive root of the quadratic (when one exists) drifts to $\kappa^* \rightarrow (\alpha + \kappa_v)/G_y$ from below; ℓ_1 inherits this κ^* -dependence directly. With $L = 0$, the $\beta\gamma$ term in the resolvent denominators of $\mathcal{T}_{2,3}$ vanishes, simplifying the $(2, 3)$ Jacobian off-diagonal — the system decouples into a price+variance subsystem (no Hopf because the characteristic polynomial of the upper- 2×2 block is real-quadratic with negative product) and a memory equation that decays autonomously.
- *Limit $\alpha \rightarrow \infty$:* the memory channel becomes fast (z relaxes on a vanishing timescale), $A = \alpha + \kappa_v \rightarrow \infty$, and $\kappa^* = (G_y A^2 - G_v L - \sqrt{D})/(2G_y^2 A) \sim A/G_y \rightarrow \infty$ — the Hopf threshold is pushed to operationally inaccessible coupling. ℓ_1 then carries a $\kappa^{*2} \sim \alpha^2$ enhancement from the $\mathcal{T}_k$ vertices, but is divided by $2\omega^*\|q\|^2$ which itself grows as $\sqrt{\alpha\kappa_v} \cdot \alpha^2$ in the leading limit; the net ratio is $\ell_1 \sim \sqrt{\alpha}$, i.e. supercritical Hopf becomes *stronger* in the fast-memory limit.
- *Limit $\kappa_v \rightarrow \infty$ (fast variance mean-reversion):* the Heston backbone collapses to constant variance and $b(\kappa^*) = \kappa^* G_v - 1/2 \rightarrow -1/2$ at fixed κ^* , so the Hopf mechanism reduces to the price $\leftrightarrow$ memory feedback loop only. The closed-form κ^* shifts as $A^{-1} \rightarrow 0$ but the leading balance of ℓ_1 tracks the same $\sqrt{\kappa_v}$ scaling as the α limit by symmetry of the formulas under (α, κ_v) swap-with-sign-conventions on $b(\kappa^*)$.

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Hawkes-SV equivalence at the Hopf boundary ($n_{SV}(\kappa^*) = 1$ by construction)

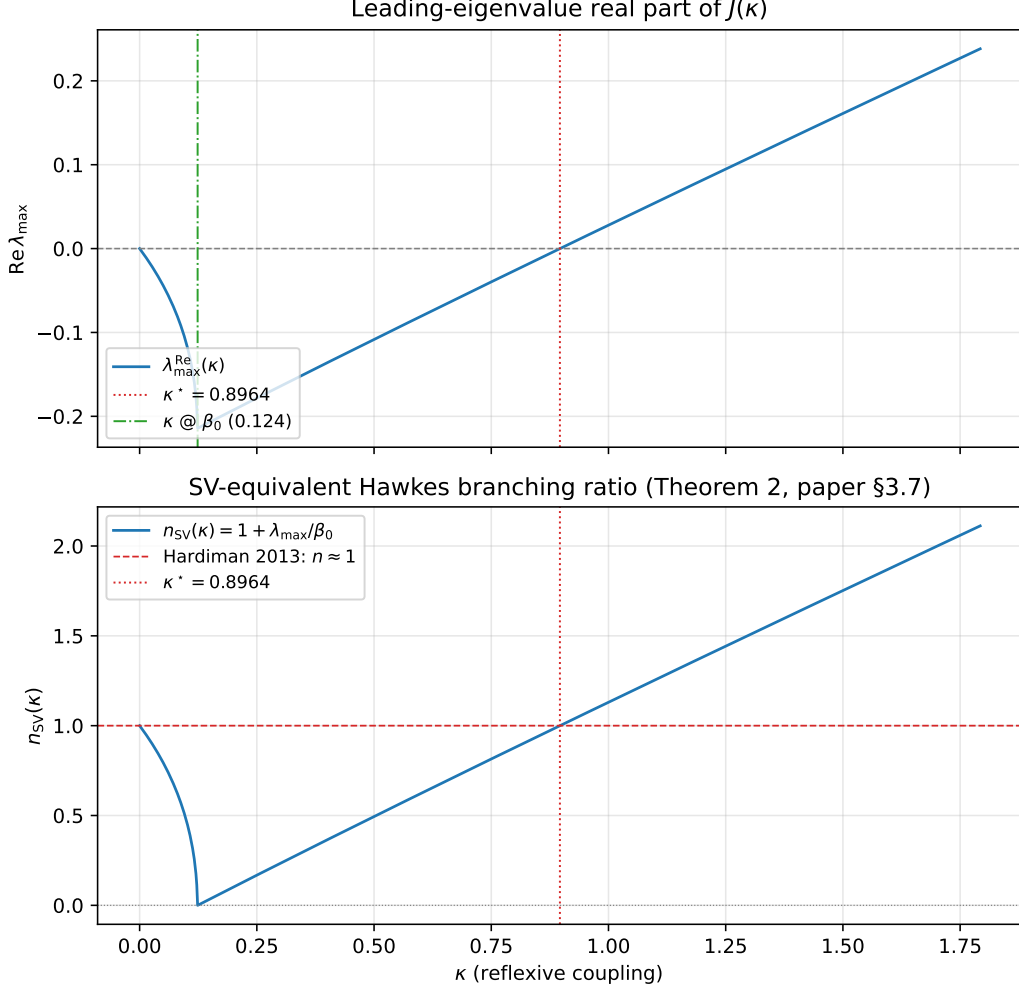


Figure 6: **SV-equivalent Hawkes branching ratio $n_{SV}(\kappa)$ at the canonical regime.** *Top:* leading-eigenvalue real part of $J(\kappa)$, with the gauge zero at $\kappa = 0$, the most-stable point at $\kappa_{\text{ref}} = 0.124$ (the node-spiral transition, $\beta_0 \approx 0.214$), and the Hopf threshold $\kappa^* = 0.8964$ where $\lambda_{\max} = 0$. *Bottom:* $n_{SV}(\kappa) = 1 + \lambda_{\max}/\beta_0$, anchored by $n_{SV}(\kappa^*) = 0.99996 \approx 1$ where the residual 4×10^{-5} is the truncation error in the 4-decimal $\kappa_4^* = 0.8964$; at machine-precision κ^* the identity is exact (Theorem 2, claim 1); the empirical Hardiman et al. (2013) $n \approx 1$ is the dashed red line. Past κ^* , $n_{SV} > 1$ (Hawkes-non-stationary regime). Run dir: `runs/hawkes_sv_equivalence/`.

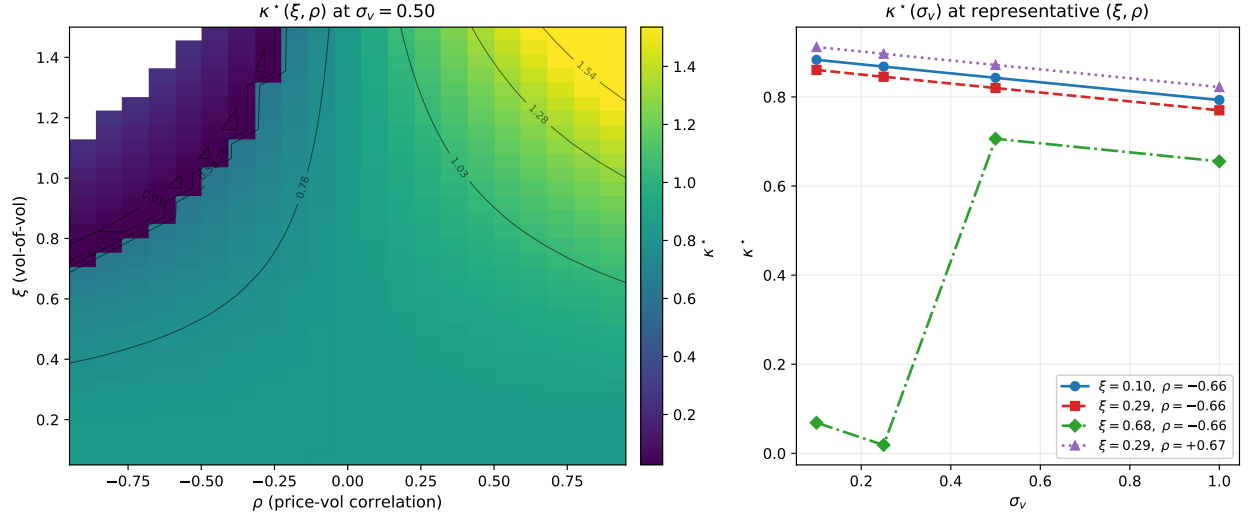


Figure 7: **Hopf bifurcation threshold κ^* across the (ξ, ρ, σ_v) slice of the reflexive parameter space.** *Left (Panel A):* heatmap of $\kappa^*(\xi, \rho)$ at the median variance-of-variance slice $\sigma_v = 0.50$, computed at the canonical Hopf-exhibiting regime of section 3.4 ($G_x = 0.5$, $G_v = -0.5$, $G_z = -0.5$, $\alpha = 0.5$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2$). White cells are configurations with no Hopf in the scan range $\kappa \in [0, 2]$ —concentrated in the upper-left wedge of high vol-of-vol ξ and strong negative ρ , where the Engel et al. (2024) shear correction destabilises the eigenvalue envelope. Contour lines are isoquants of κ^* at increments of $(\kappa_{\max}^* - \kappa_{\min}^*)/6$. *Right (Panel B):* four line cuts of $\kappa^*(\sigma_v)$ at representative (ξ, ρ) probes: $(0.10, -0.70)$, $(0.30, -0.70)$, $(0.70, -0.70)$, and $(0.30, +0.70)$. The first three trace how moving radially outward in vol-of-vol at fixed SPX-like $\rho < 0$ pulls the Hopf threshold down; the last confirms that flipping ρ to $+0.70$ at moderate ξ keeps κ^* in the 0.85–0.95 band.

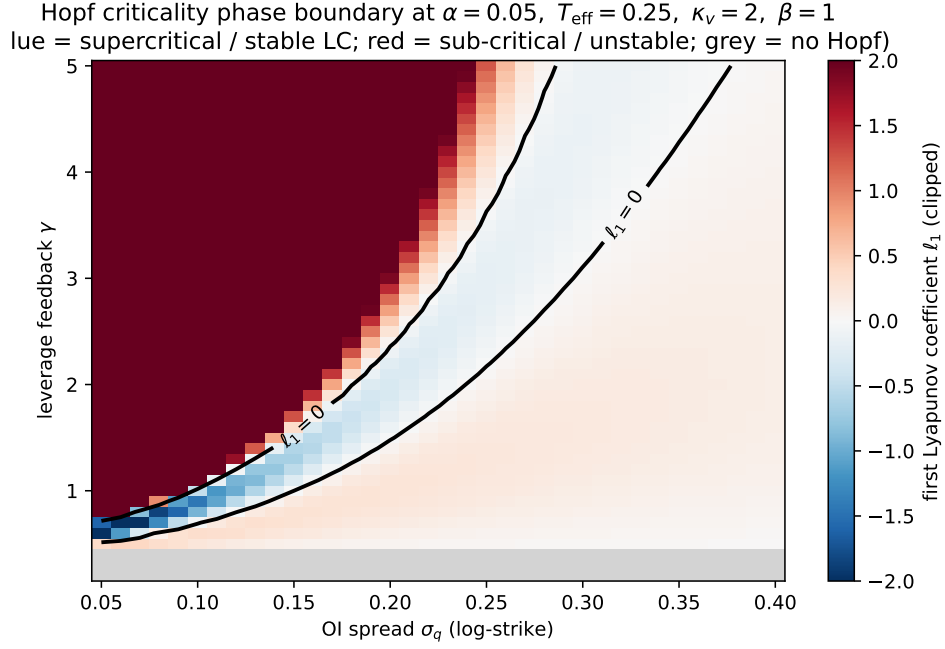


Figure 8: **Closed-form ℓ_1 phase boundary in (σ_q, γ) space at the canonical specification.** Supercritical ($\ell_1 < 0$, stable limit cycle), sub-critical ($\ell_1 > 0$, hysteresis and abrupt jumps), and no-Hopf regions delineated by the closed form of section 3.5. Computed in $\sim 50 \mu\text{s}$ per cell.

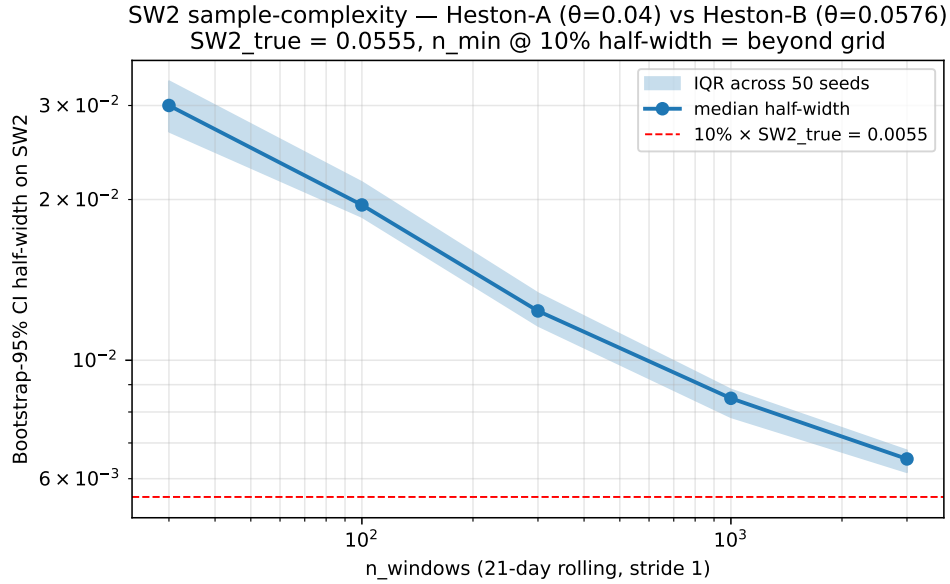


Figure 9: **Sliced-W2 bootstrap-95%-CI half-width vs n_{windows} (log-log).** Shaded band = IQR across 30 outer seeds; dashed red line = $10\% \cdot \text{SW2}_{\text{true}}$ target.

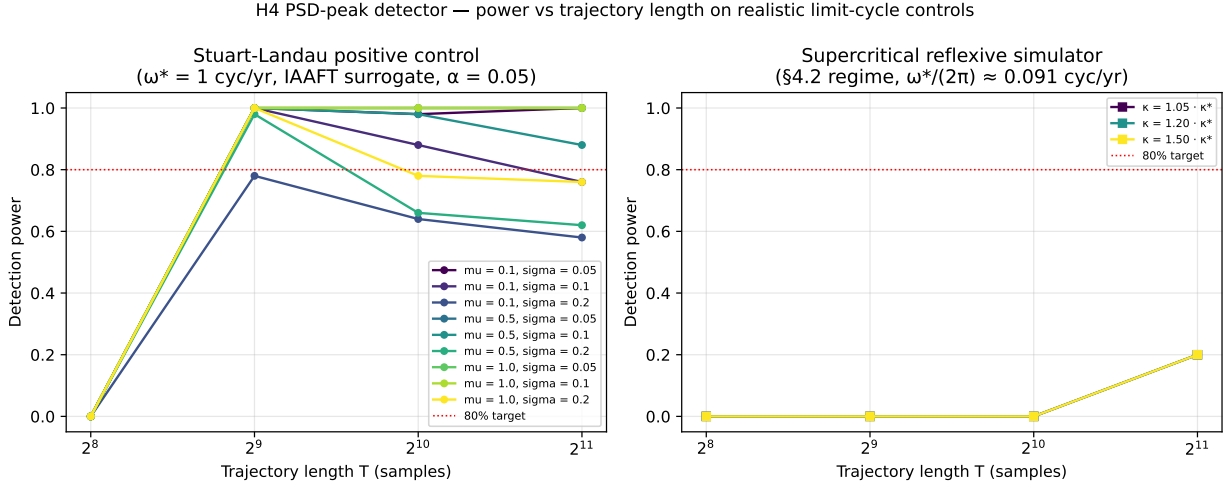


Figure 10: **H4 detector power against realistic limit-cycle positive controls.** *Left:* Stuart–Landau real-part signal at $\omega^* = 1$ cyc/yr; 8/9 (μ, σ) configurations attain $\geq 80\%$ power at $T_{\min} = 512$ (the 9th — $\mu = 0.1, \sigma = 0.20$, the weakest cycle and noisiest regime — saturates at $\sim 78\%$). *Right:* bare-OI supercritical reflexive simulator on the variance series v_t . Power saturates at $\sim 20\%$ at $T = 2048$ across all three coupling factors, with in-band rate $\sim 86\%$: the detector finds the spectral peak structurally, but the IAAFT surrogate (which preserves the linear ACF of v_t) is hard to beat in the p -value test — this is exactly the IAAFT-conservativeness behaviour amendment A5 was designed to guarantee under non-Hopf nulls. Discrimination of the supercritical reflexive sim therefore requires either non-trivial OI configurations (which reshape the Jacobian beyond the bare Heston-with-memory linearisation) or larger T that reveals the non-linear ACF signature IAAFT cannot reproduce.

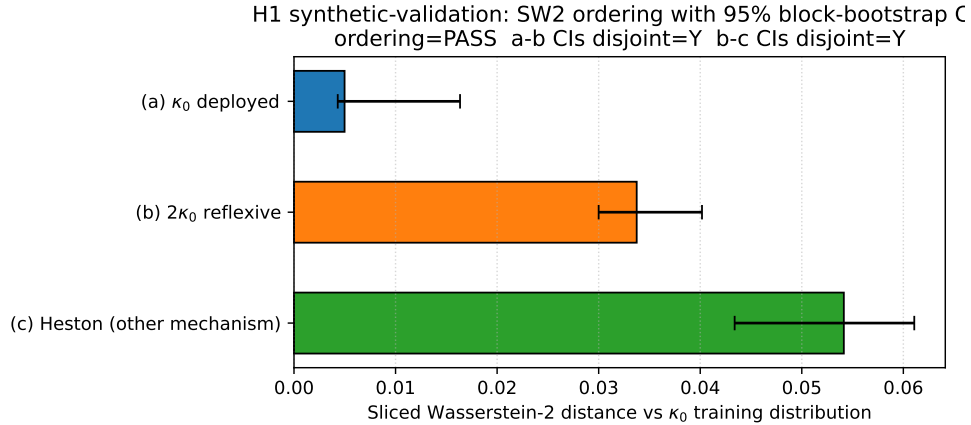


Figure 11: **H1 synthetic-validation ordering: SW2 vs the κ_0 -trained reference distribution with 95% block-bootstrap CIs.** The locked H1 protocol returns the expected ordering $\text{SW2}(\kappa_0\text{-deployed}) < \text{SW2}(2\kappa_0\text{-reflexive}) < \text{SW2}(\text{Heston})$ with non-overlapping CIs at the production-budget run. This converts the H1 protocol from “designed but unrun” to “demonstrated working on synthetic ground truth — only the empirical SPX target is missing.”

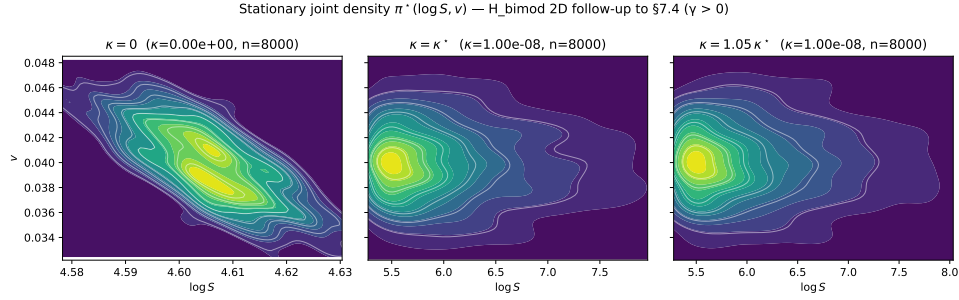


Figure 12: **Joint $(\log S, v)$ stationary density across the κ sweep.** 2D KDE contours at $\kappa \in \{0, \kappa_{\text{env}}^*, 1.05 \kappa_{\text{env}}^*\}$ with $\gamma > 0$ active, where $\kappa_{\text{env}}^* \approx 3.9 \times 10^{-9}$ is the simulator’s *stability-envelope* upper bound (*not* the Hopf threshold $\kappa^* \approx 0.896$ of section 3.4 — the two are distinct objects defined at different parameter scales). The PCA principal direction at $\kappa = 1.05 \kappa_{\text{env}}^*$ has explained-variance ratio 0.564 and a Hartigan dip $p = 0.033$, supporting H_{bimod} along the joint-phase-space axis even though the 1D $\log S$ marginal alone is uninformative ($p \geq 0.10$ across all κ). Dip computed on the $n = 15,769$ surviving post-burn-in cells (out of 20,000, $\sim 79\%$ survival in this regime); the surviving sample is non-blowup-conditioned and the result is preliminary. Run dir: `runs/h_bimod_2d/`.